

Practical session: Maths & Stats practical

EMBO Practical Course

Population genomics: background and tools

Oscar Lao

THE SOCRATIC METHOD

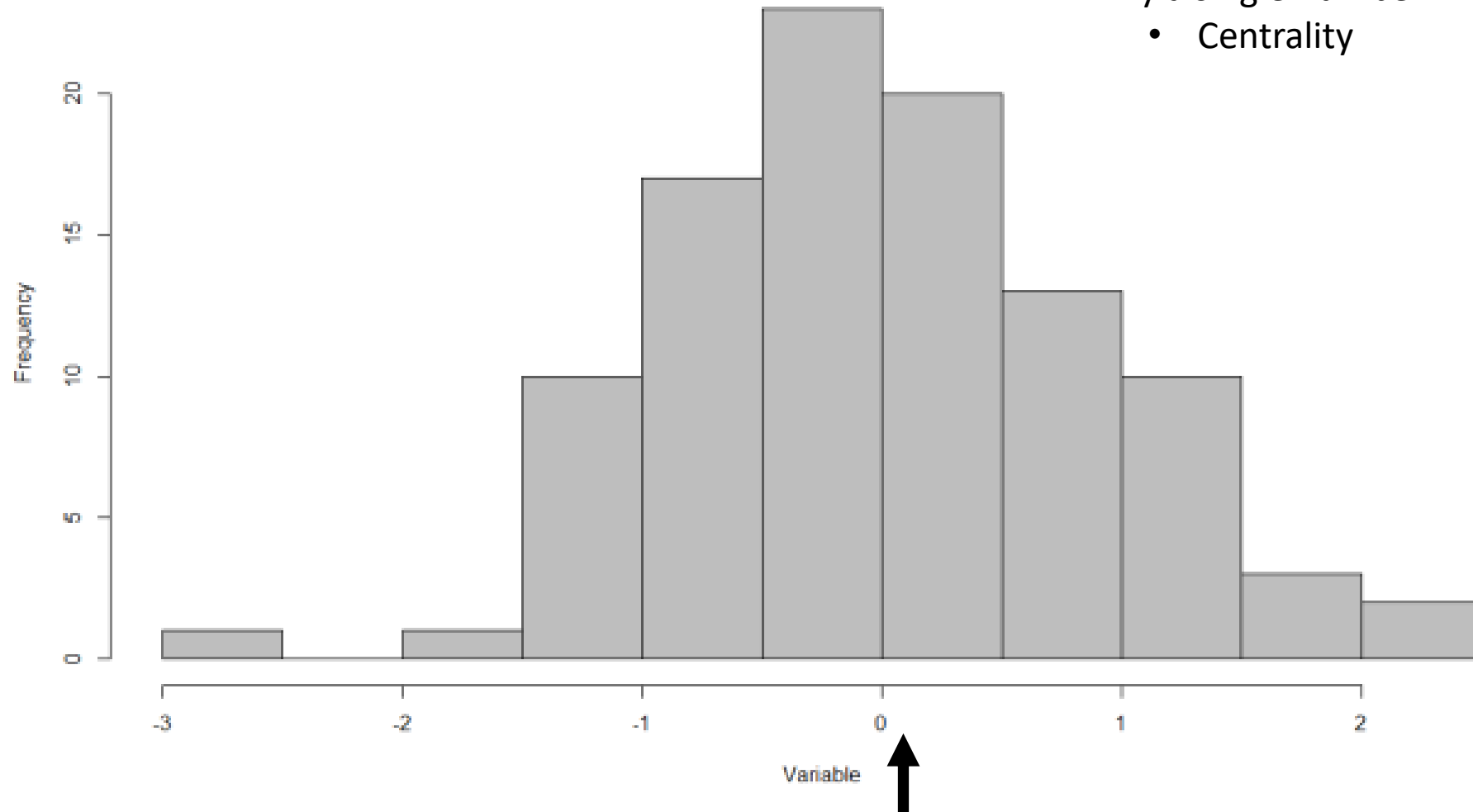


- questions are more than welcome!
- There are no wrong answers!

Distribution

How to mathematically describe a Distribution?

- By a single number:
 - Centrality



What is the mean?

What is the mean?

$$E = \sum (X - j)^2 = \sum X^2 - 2Xj + j^2$$

$$\frac{dE}{dX} = \sum 2X - 2j = 2nX - 2 \sum j$$

$$0 = \sum 2X - 2j = 2nX - 2 \sum j$$

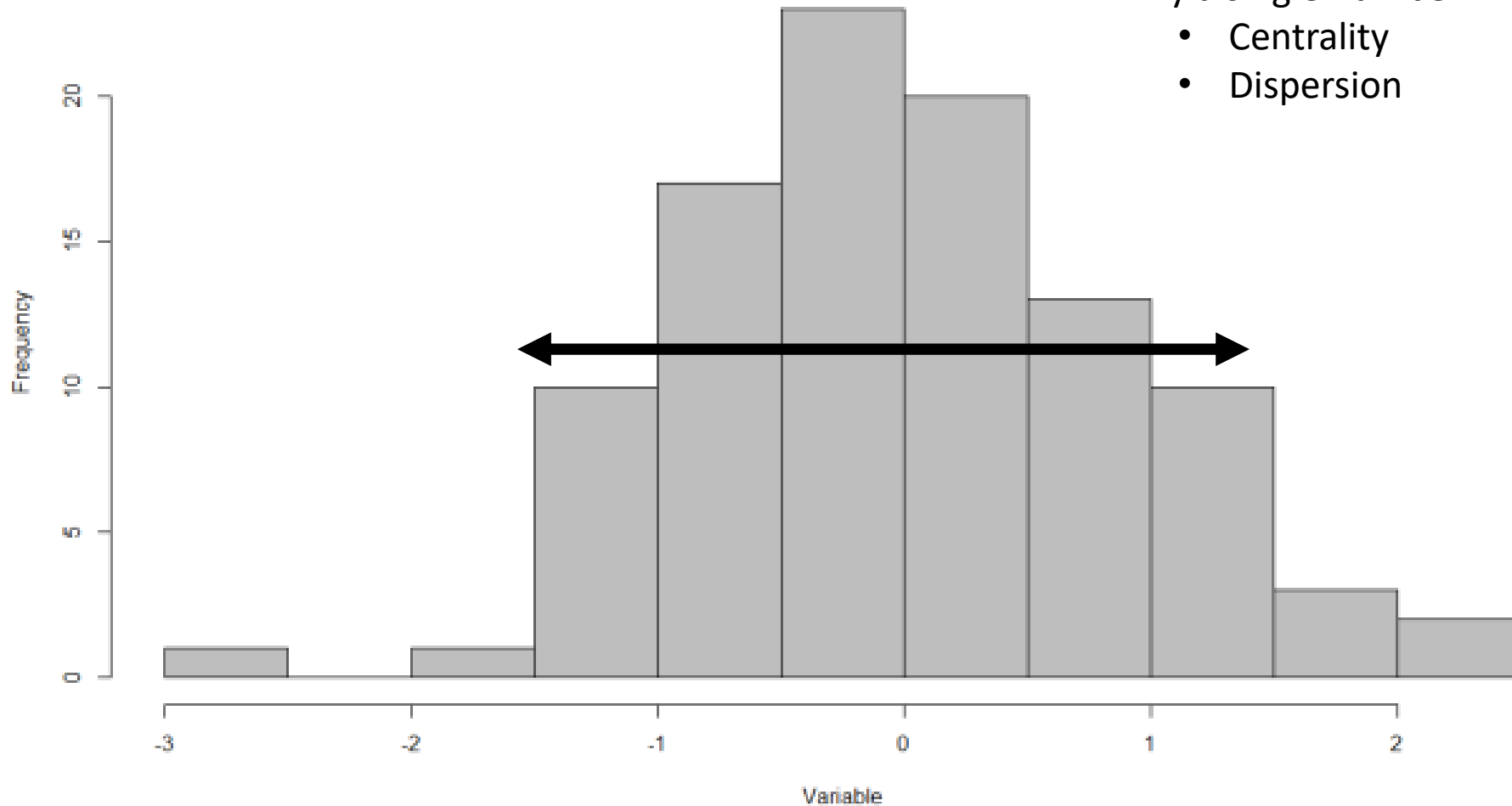
$$X = \frac{\sum j}{n}$$

“the mean is the point that minimizes the sum of squared distances to any point of the distribution”

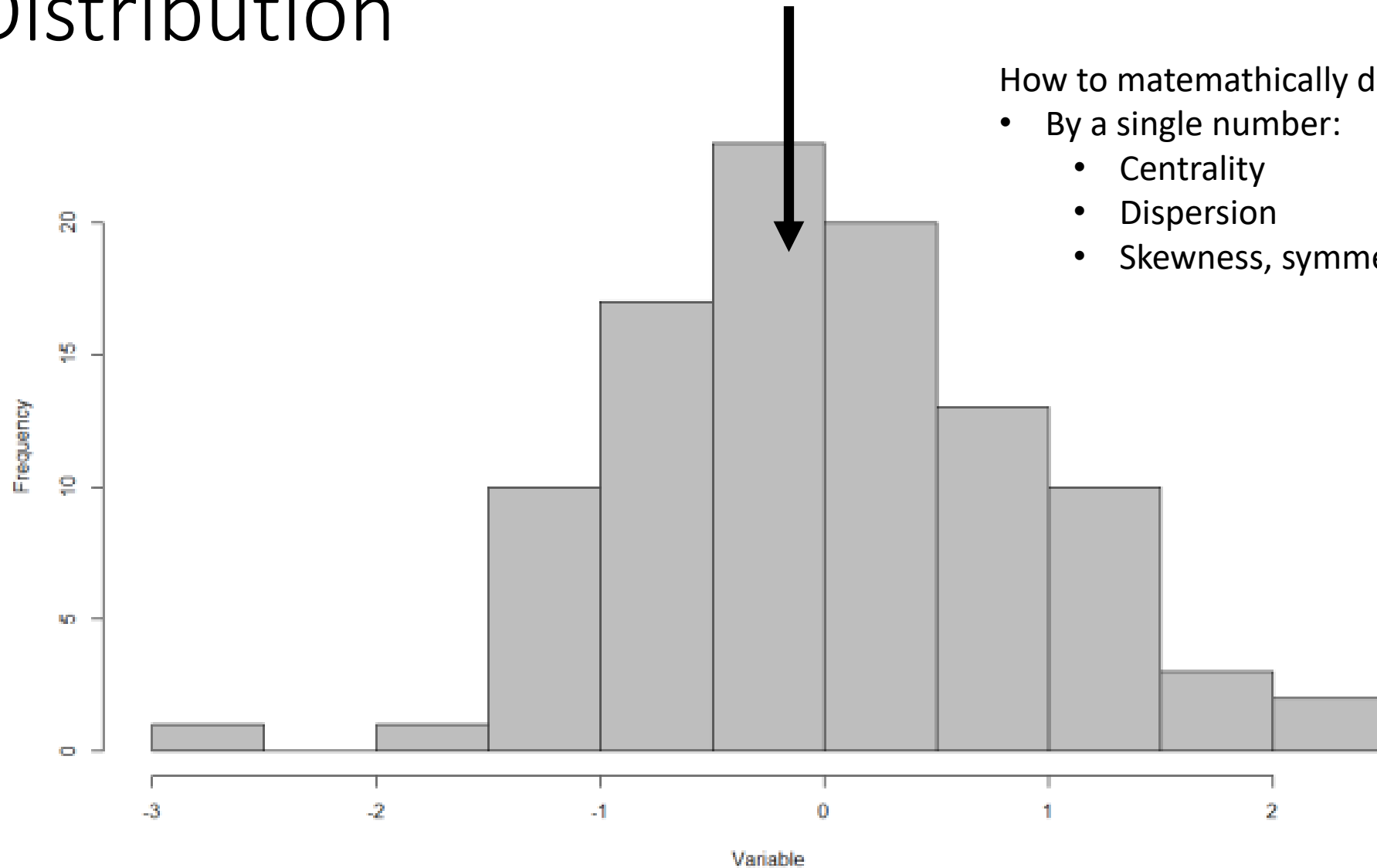
Distribution

How to mathematically describe a Distribution?

- By a single number:
 - Centrality
 - Dispersion



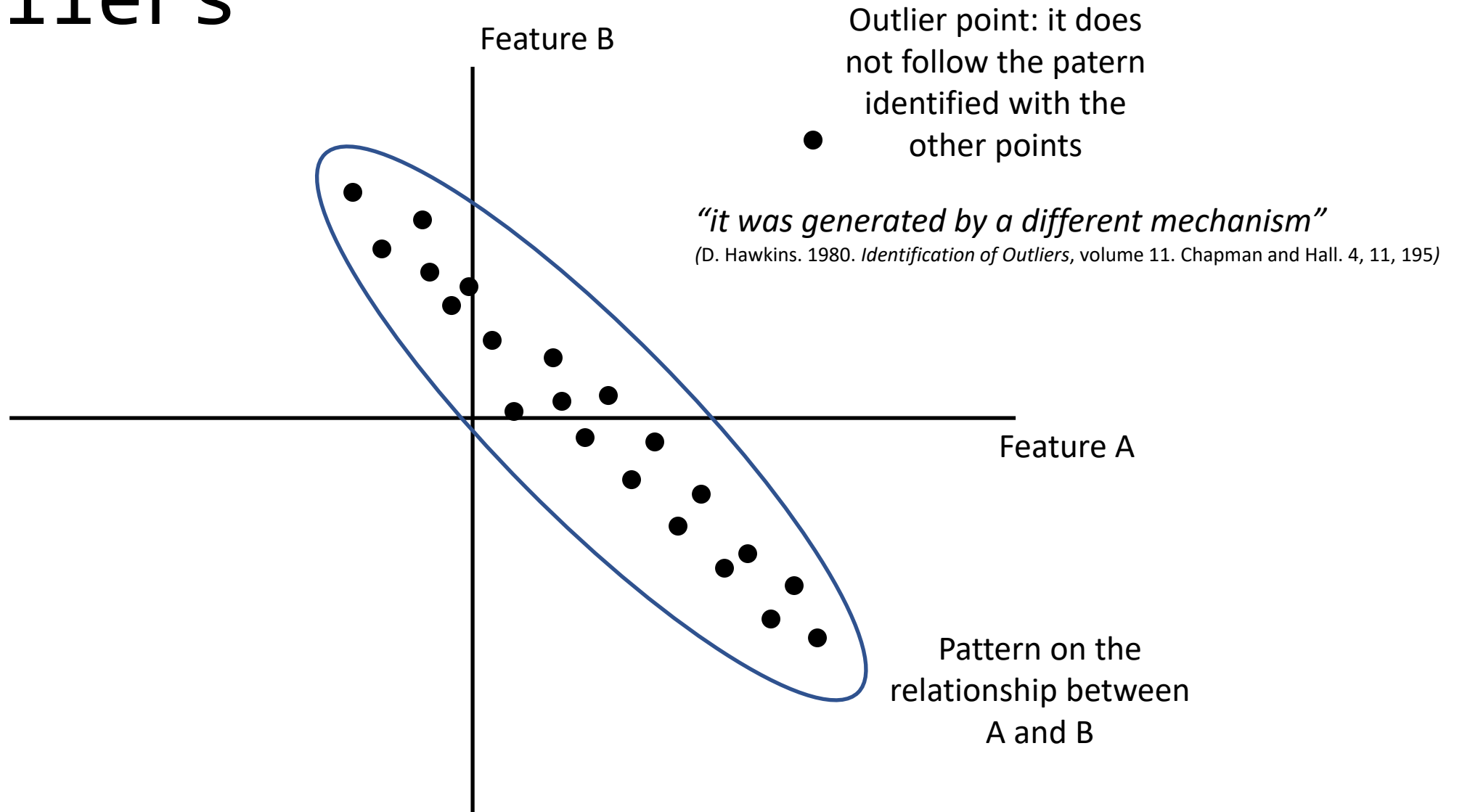
Distribution



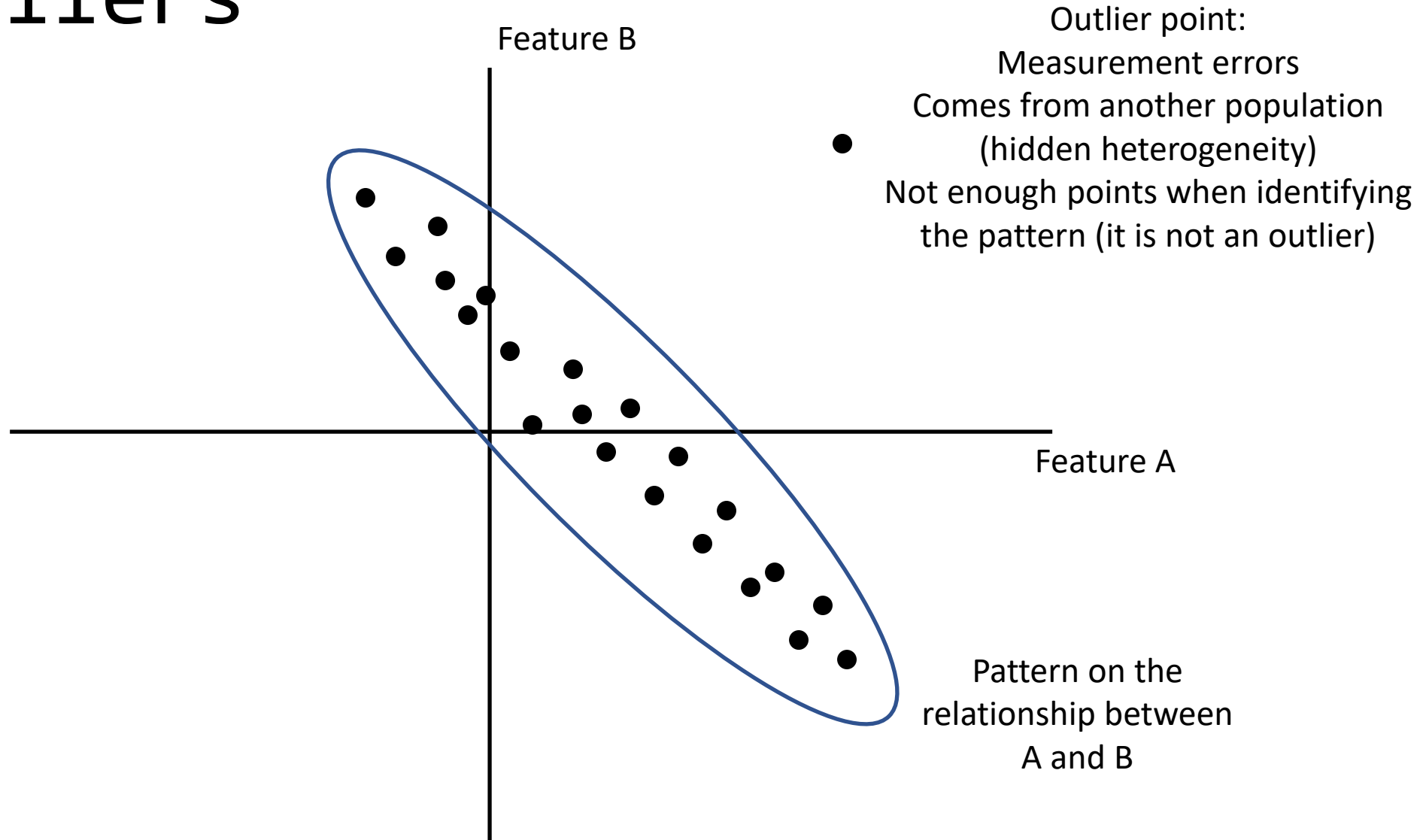
How to mathematically describe a Distribution?

- By a single number:
 - Centrality
 - Dispersion
 - Skewness, symmetry...

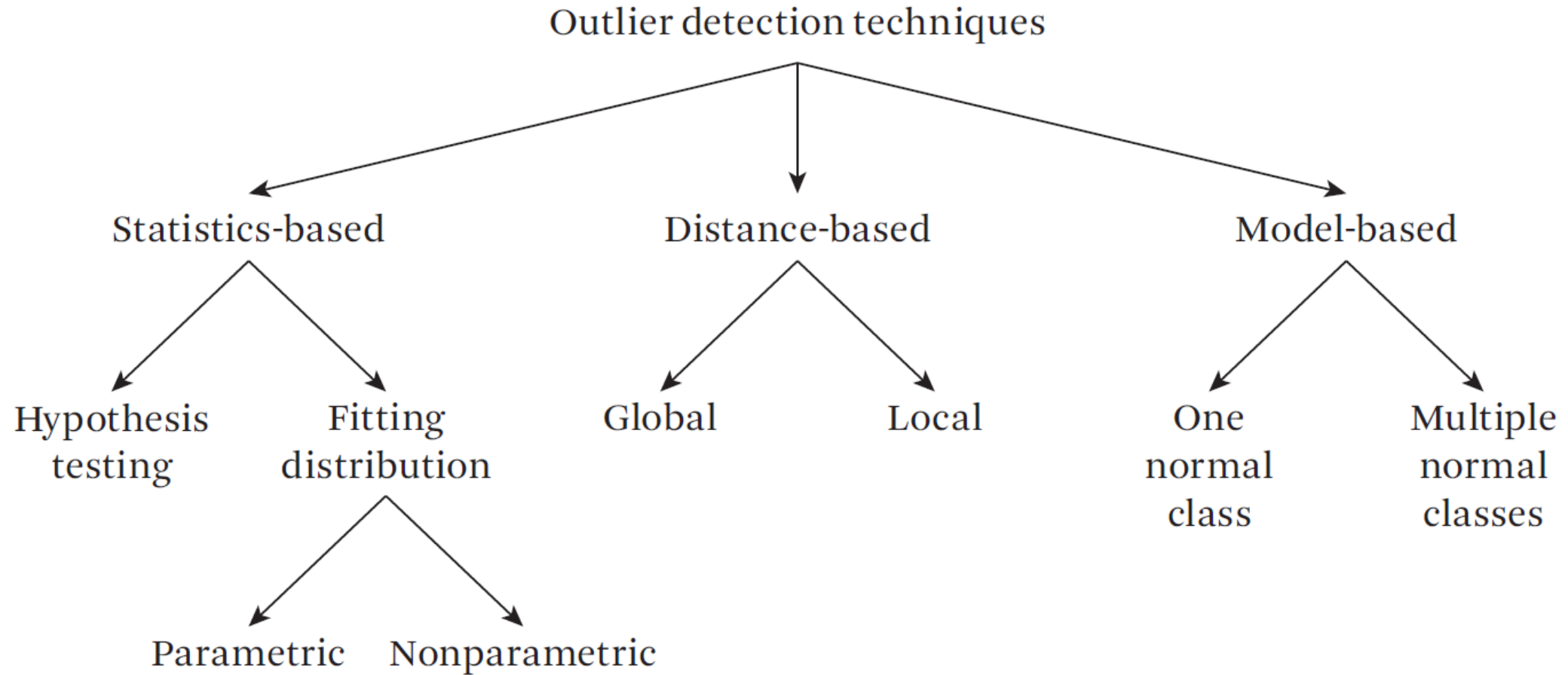
Outliers



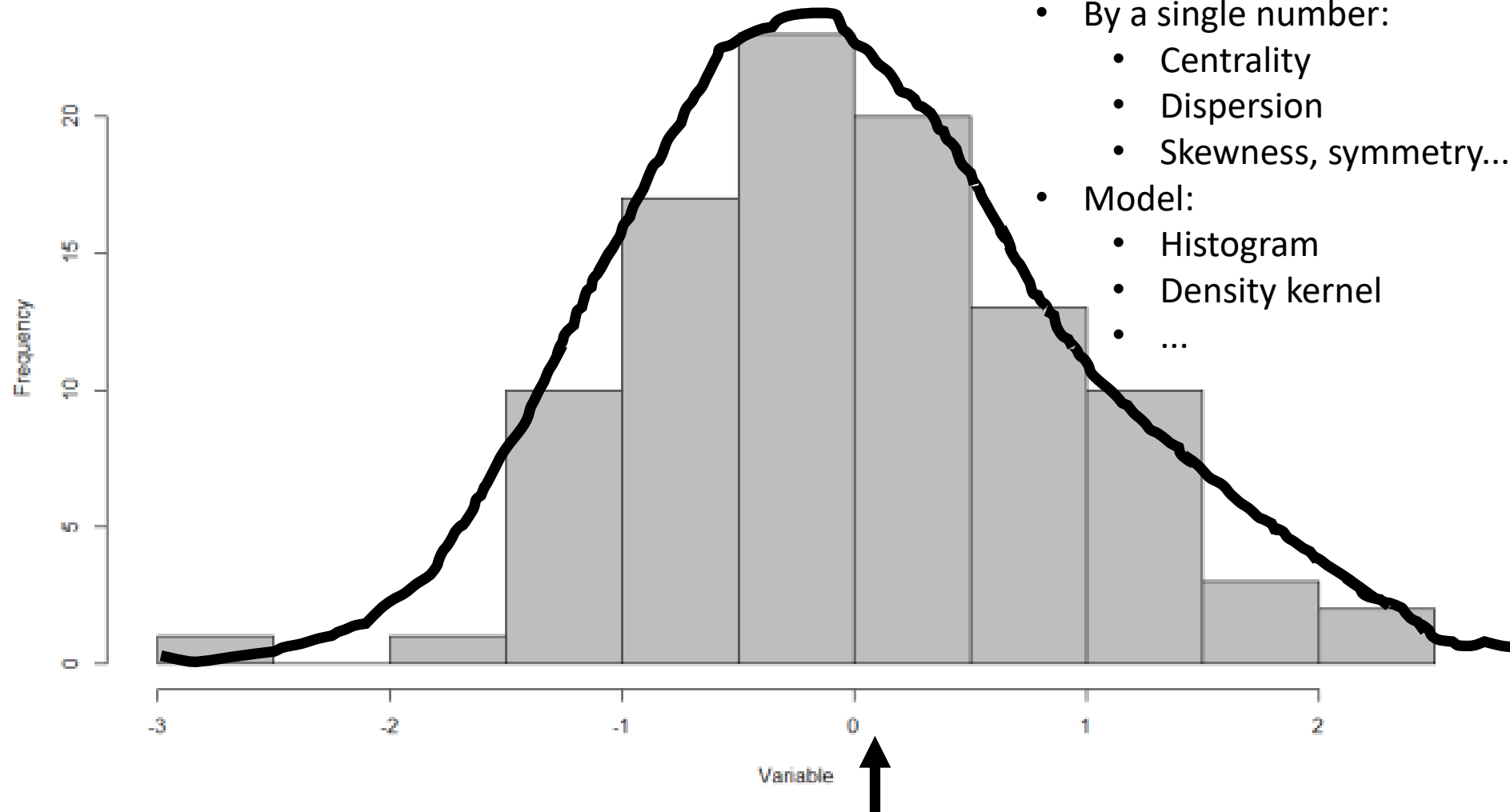
Outliers



Outlier detection



Distribution



How to mathematically describe a Distribution?

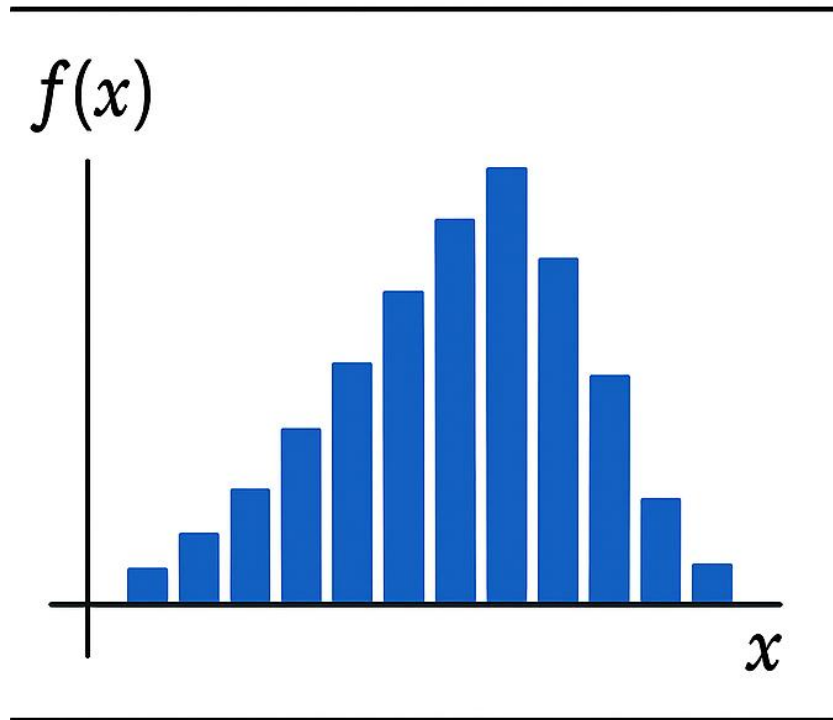
- By a single number:
 - Centrality
 - Dispersion
 - Skewness, symmetry...
- Model:
 - Histogram
 - Density kernel
 - ...

In R, exercises 1 to 3

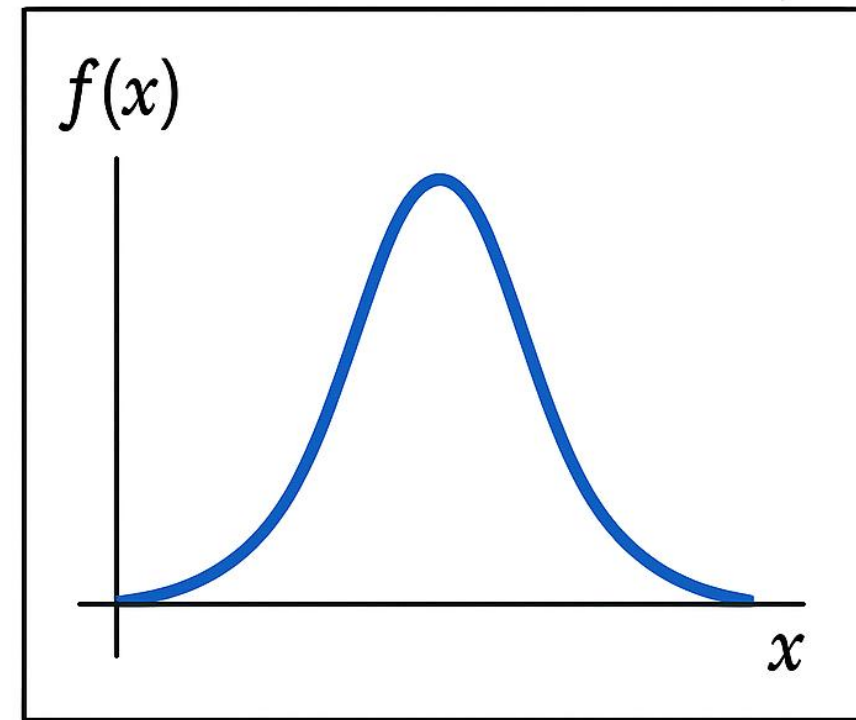
- How to describe a Distribution, robust statistics, etc

Basics of probability

Probability Distribution function

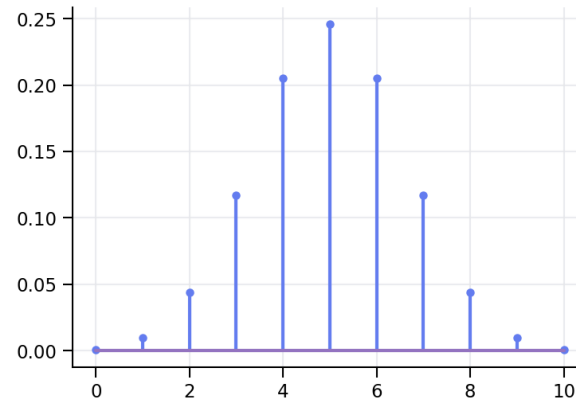


Probability Density function

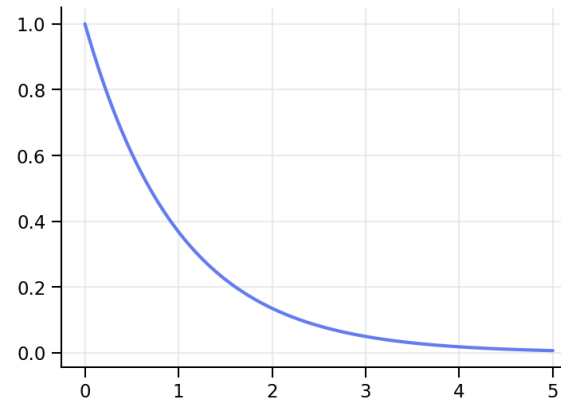


Basics of probability distributions in population genetics

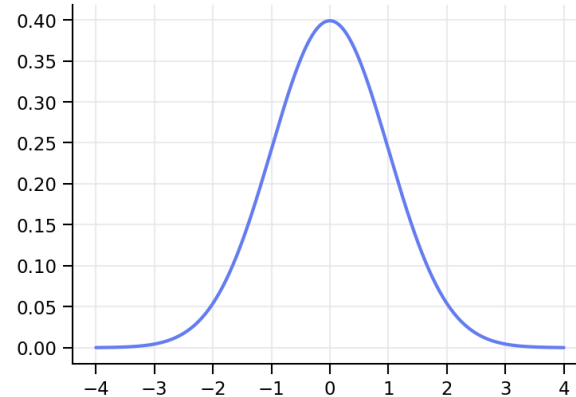
Binomial PMF ($n=10, p=0.5$)



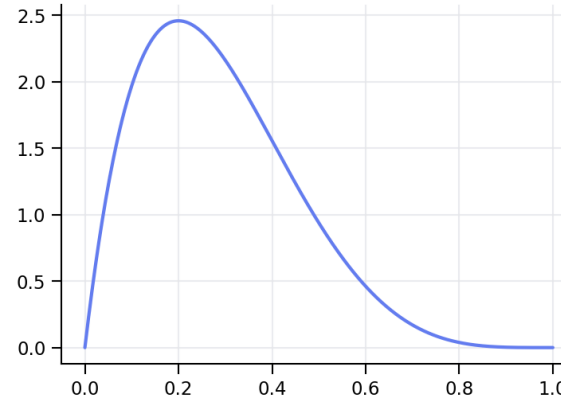
Exponential PDF ($\lambda=1$)



Normal PDF ($\mu=0, \sigma=1$)



Beta PDF ($\alpha=2, \beta=5$)



Exercise 4

- Some properties of the binomial distribution and probability concepts

The Central Limit Theorem (CLT)

- **What it does:**

The Central Limit Theorem explains the general shape of a sampling distribution.

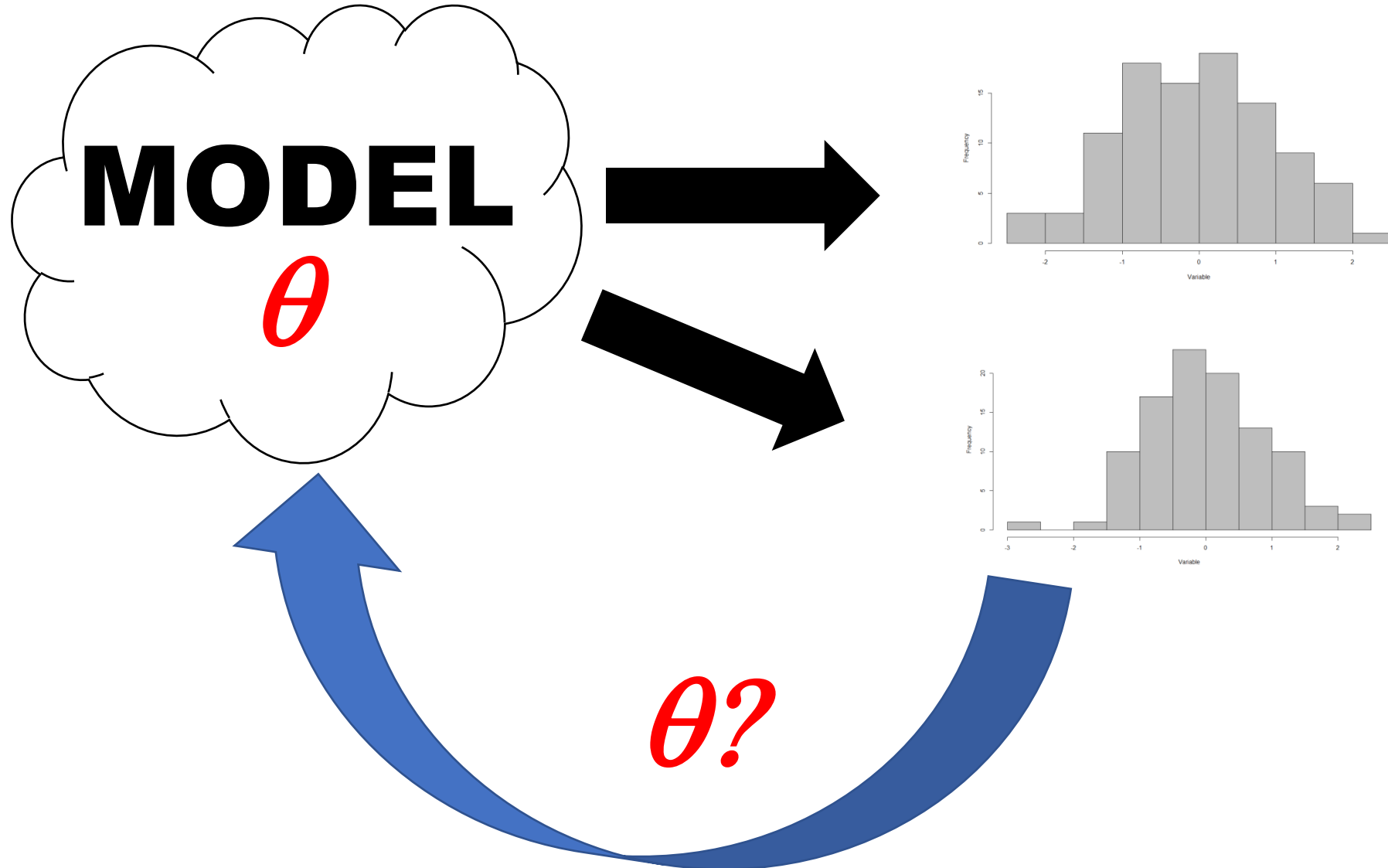
- **The concept:**

When many samples are taken from a population, the means of those samples tend to form a normal distribution (bell-shaped curve), even if the original population data is not normally distributed. The original data may be skewed, uniform, or irregular, but as the sample size increases, the distribution of the sample means becomes approximately normal. The theorem also determines the mean and standard deviation of this sampling distribution.

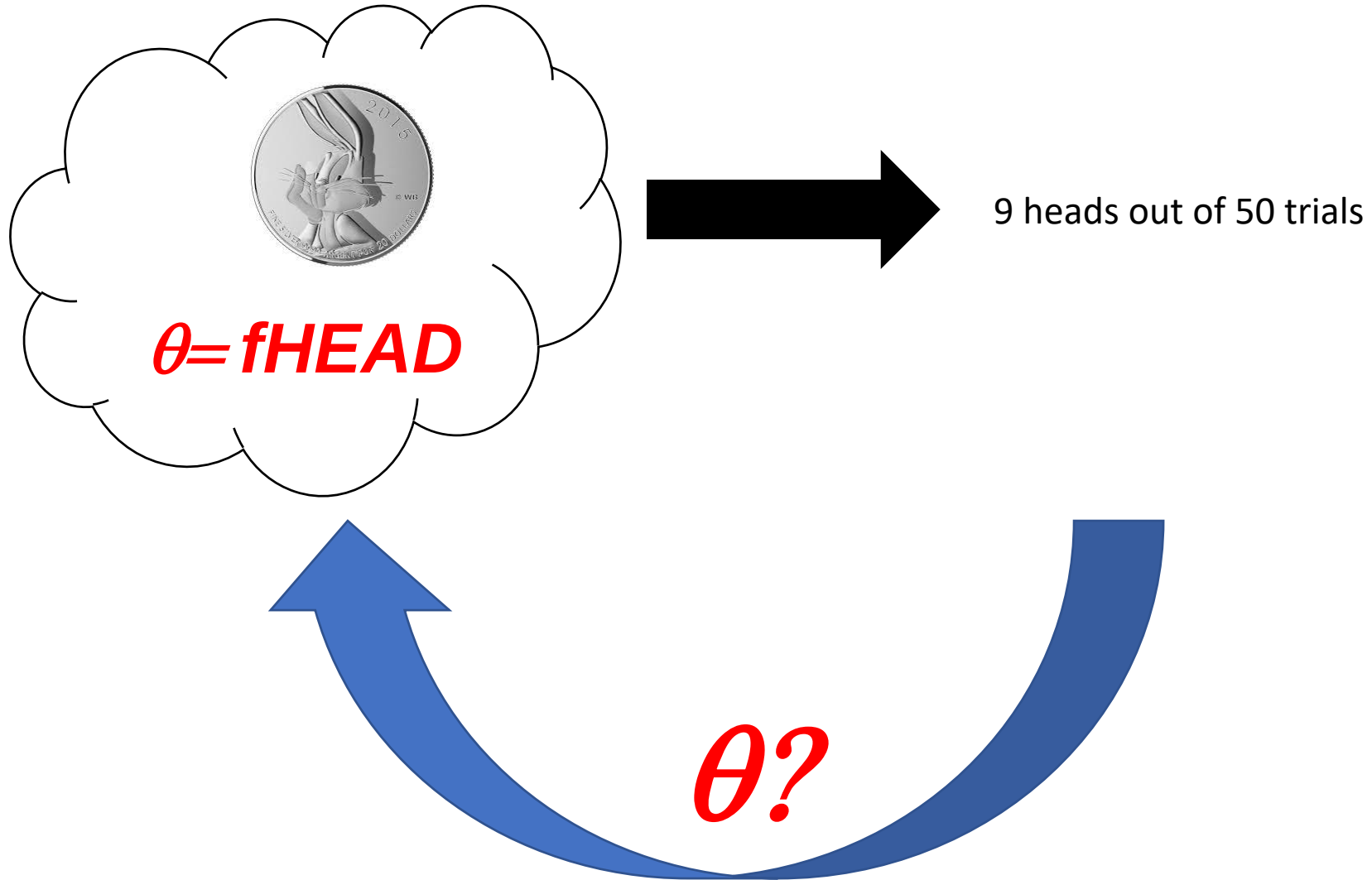
In R, exercise, 5 & 6. Empirically show the CTL

- Design an experiment in R to observe the CTL in practice.
- How can the CTL be used in population genomics?

Parameter estimation



Parameter estimation



Parameter estimation

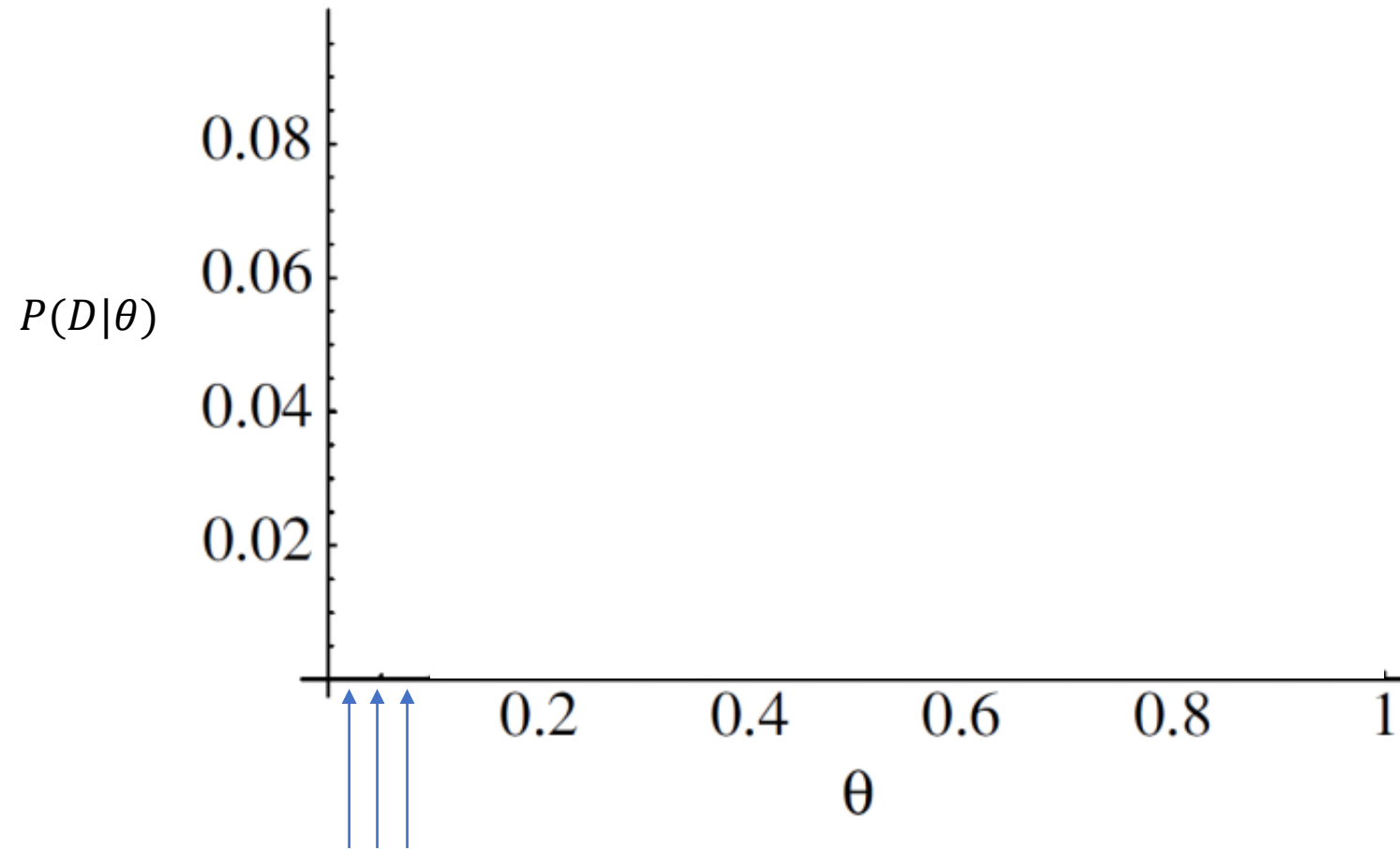


9 heads out of 50 trials

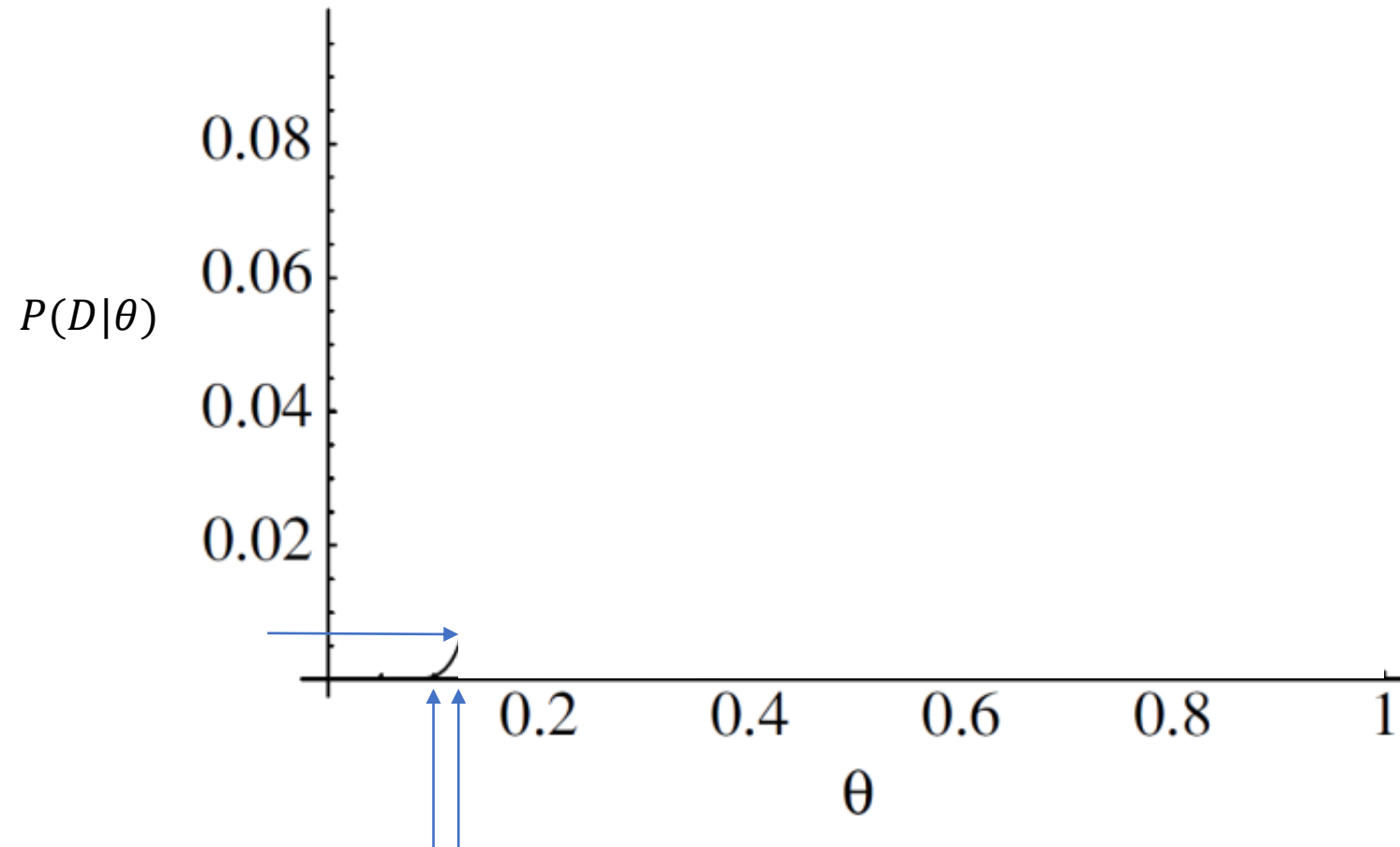
$$L(\theta) = \Pr[H = h] = \binom{n}{h} \theta^h (1 - \theta)^{n-h}$$

$\theta?$

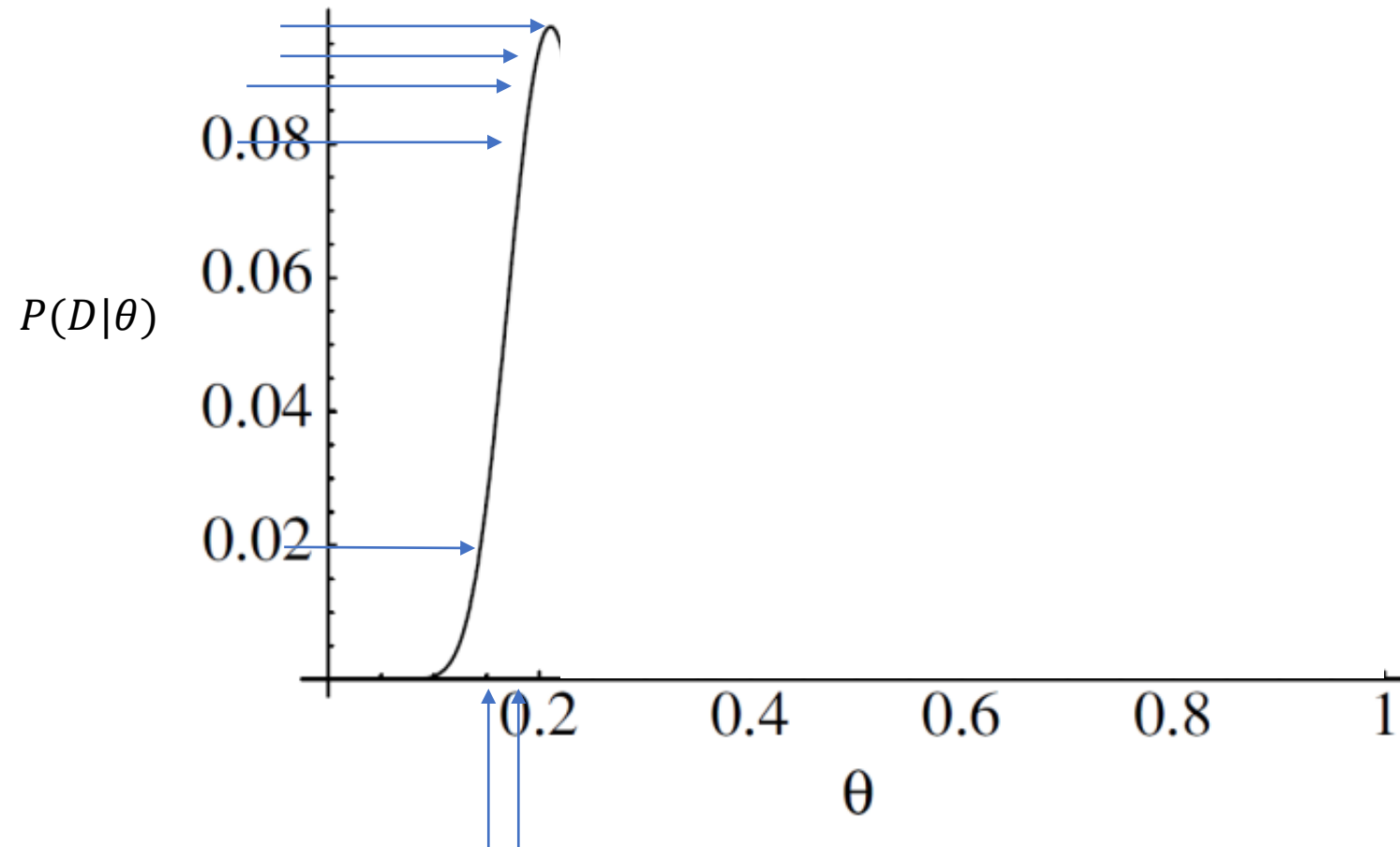
Maximum likelihood



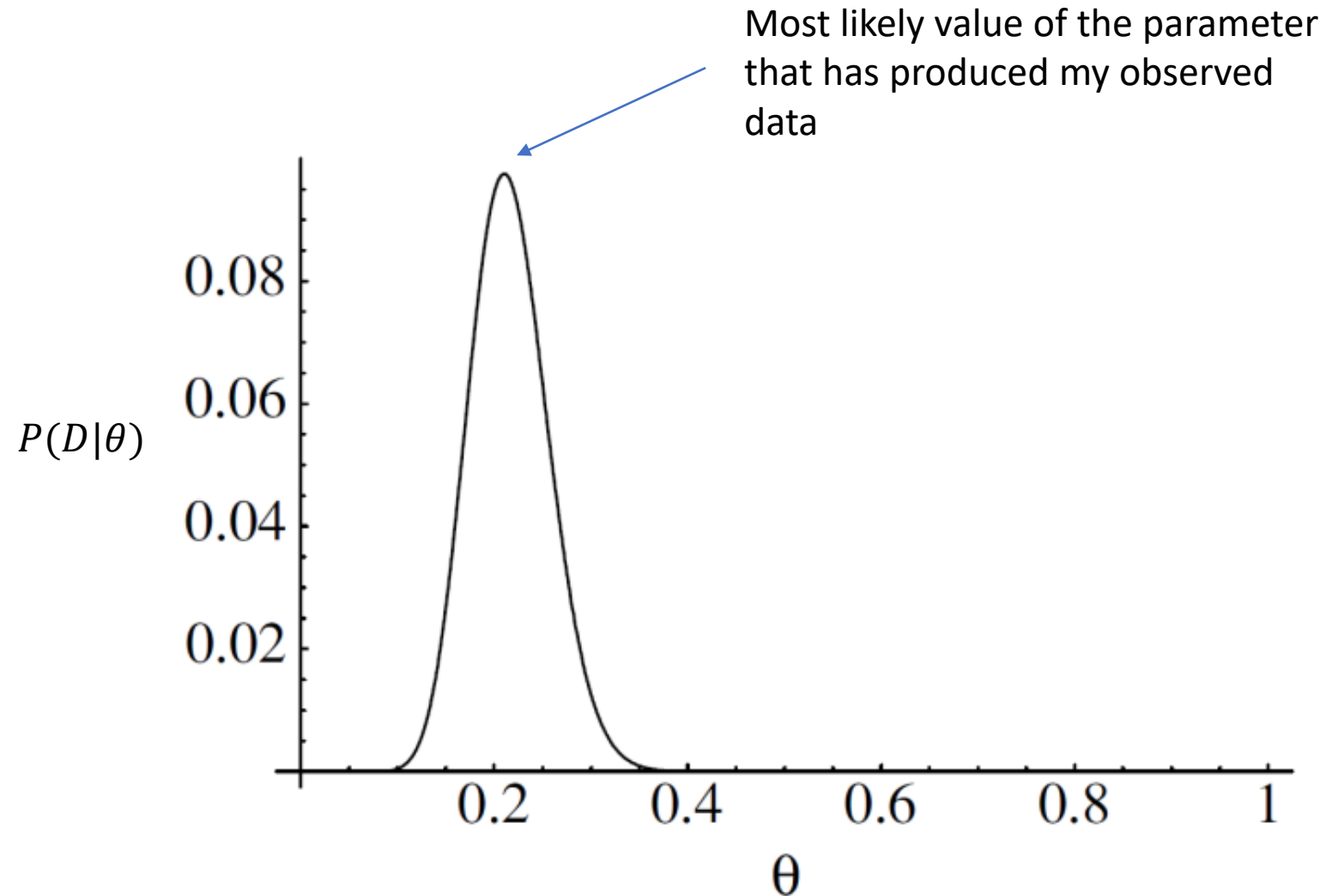
Maximum Likelihood



Maximum Likelihood



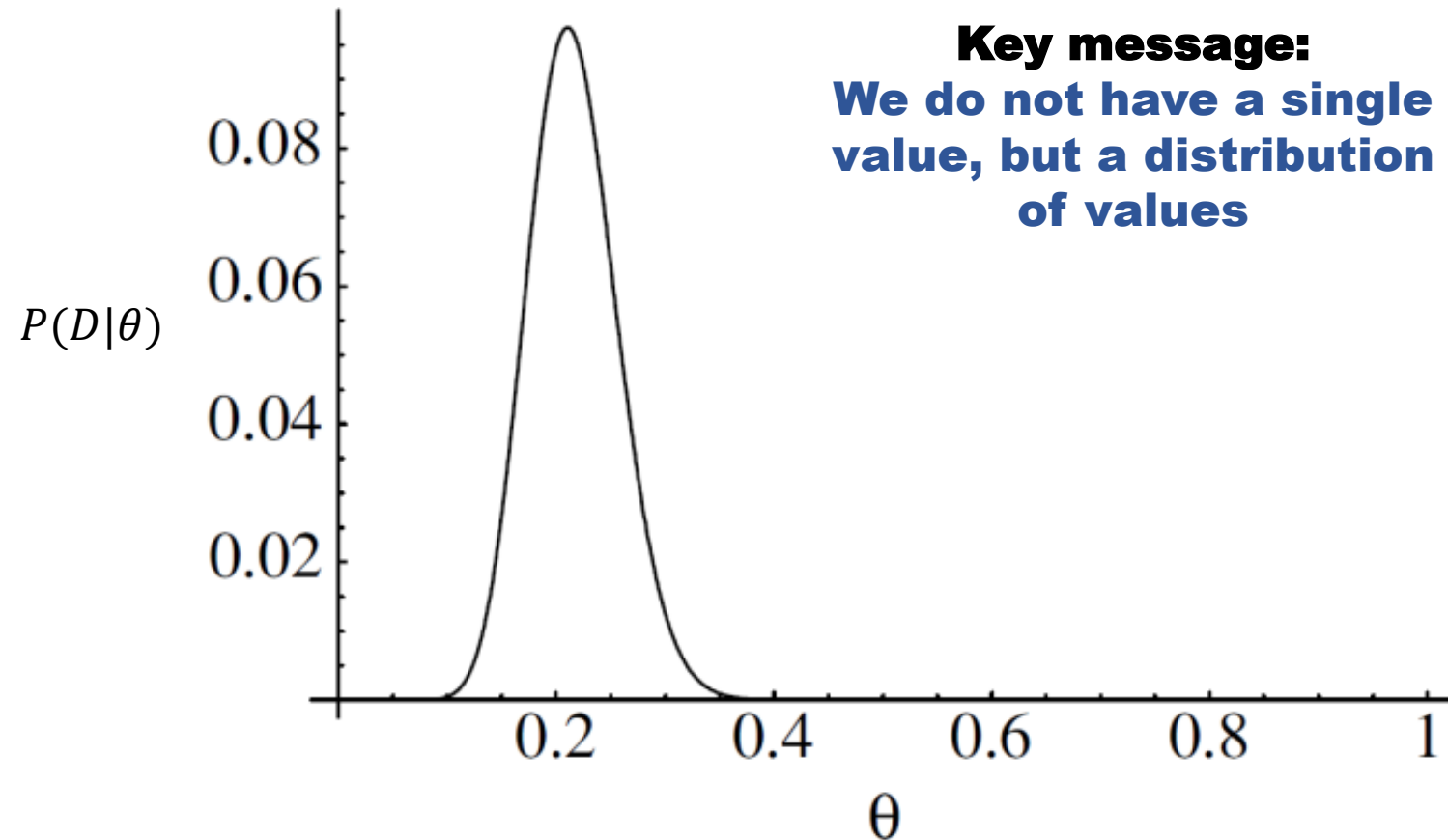
Maximum Likelihood



Exercise 7. Maximum likelihood

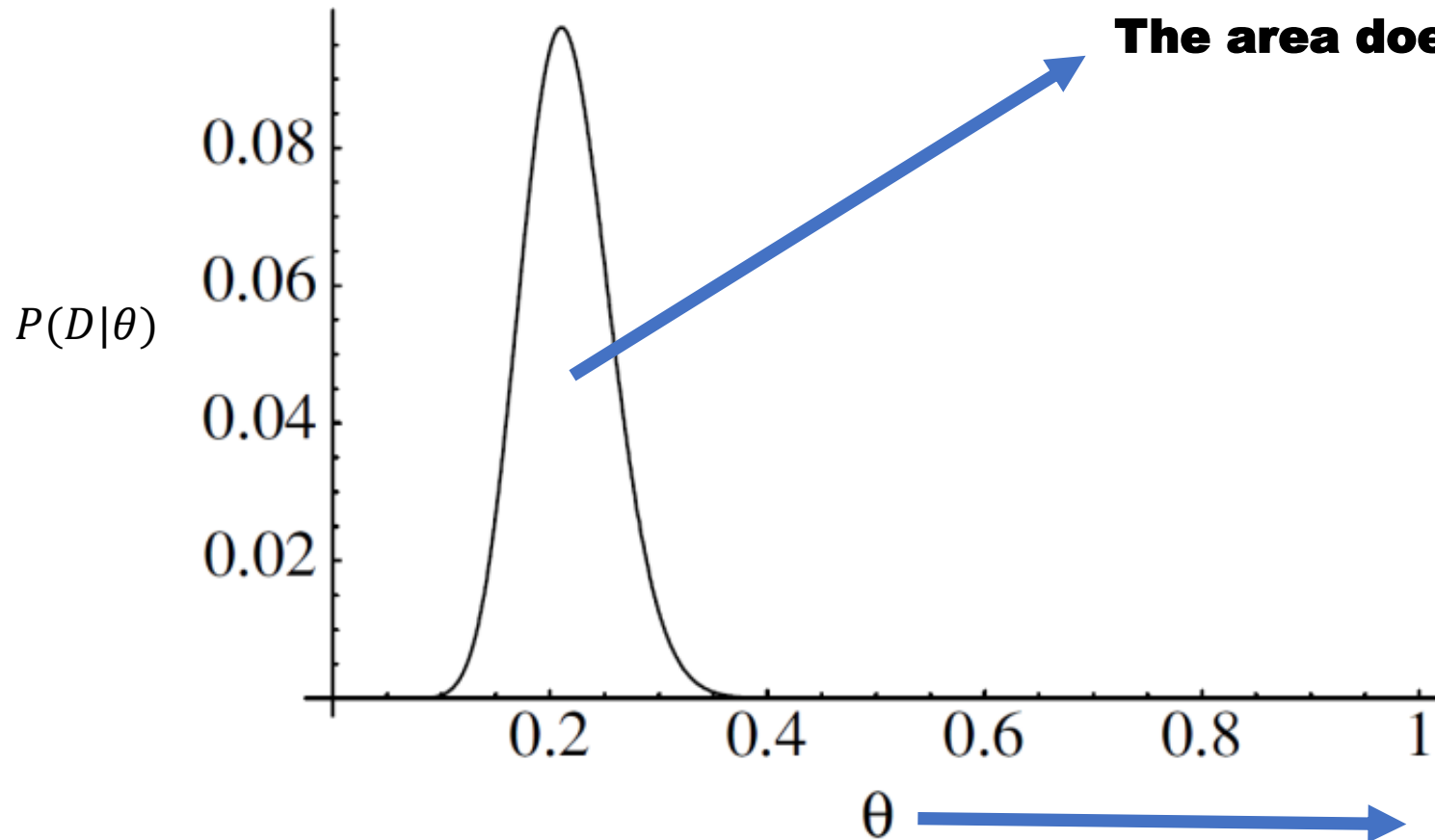
- Generate a grid of possible values and estimate the maximum likelihood value

A BAYESIAN VIEW



A BAYESIAN VIEW

$$E(\theta) = \int_0^1 P(D|\theta) * \theta d\theta$$

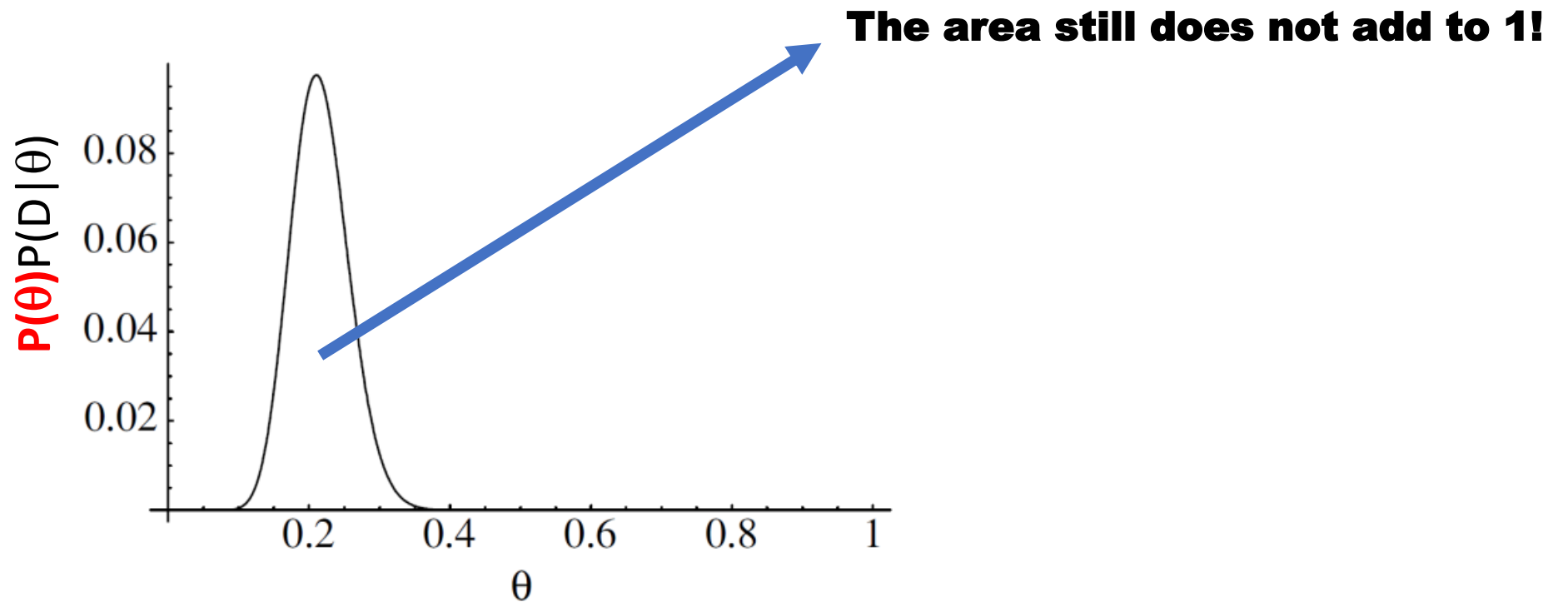


The area does not add to 1!

**What if we knew
from previous
experiences that
some values are
more likely than
others?**

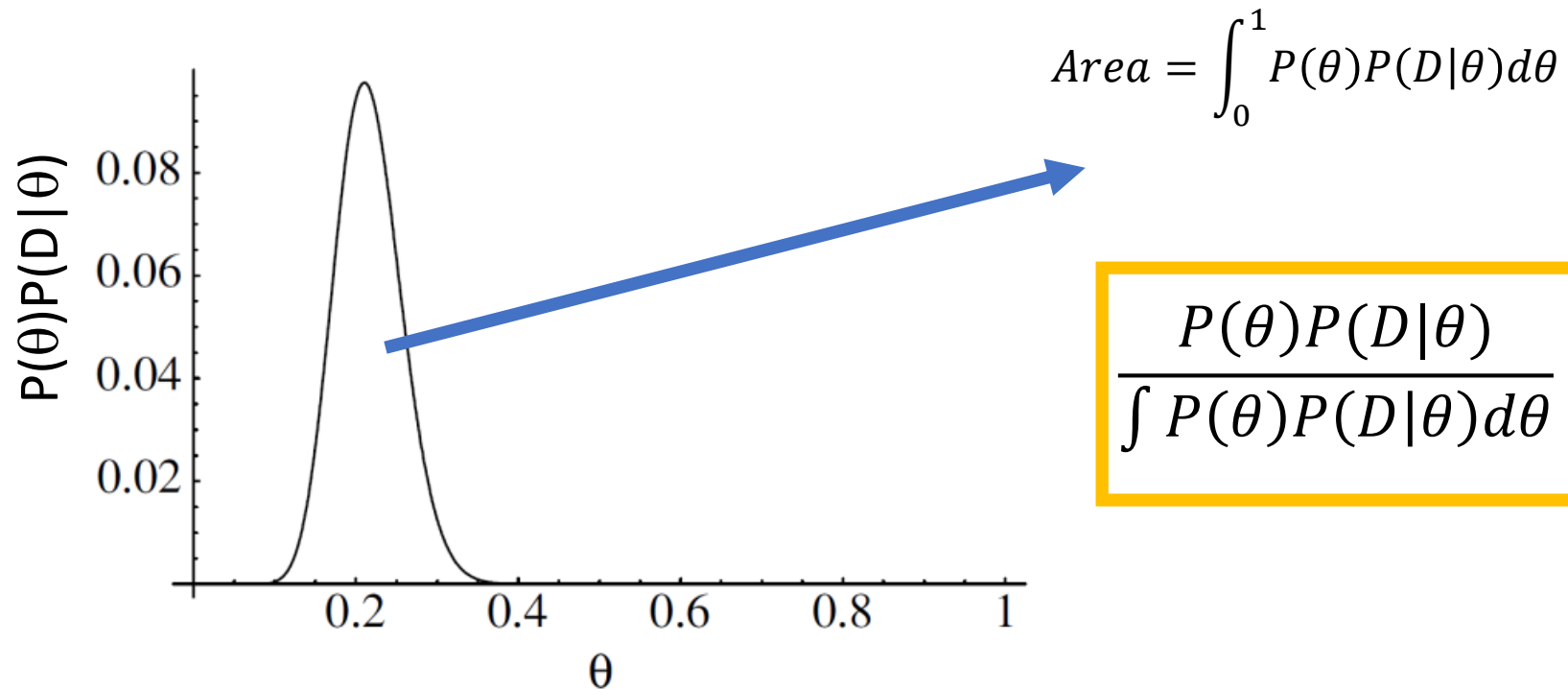
A BAYESIAN VIEW

“mixture of distributions”



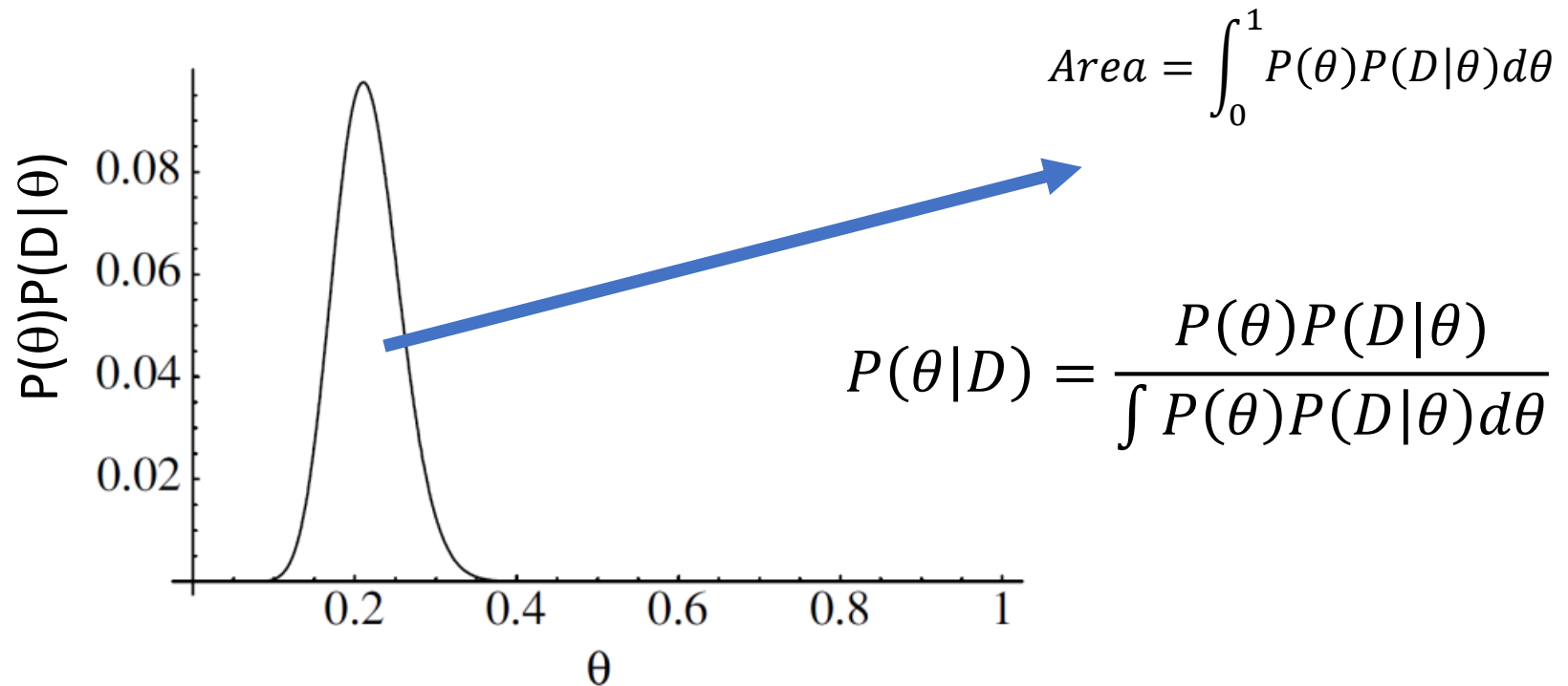
A BAYESIAN VIEW

Bayes theorem! $P(A|B) = \frac{P(A)P(B|A)}{P(B)}$



A BAYESIAN VIEW

Bayes theorem!
$$P(A|B) = \frac{P(A)P(B|A)}{P(B)}$$



A BAYESIAN VIEW

A formal way of expressing it...

$$P(D; \theta) = P(\theta)P(D|\theta) = P(D)P(\theta|D)$$

Prior knowledge $\swarrow$

$$P(\theta|D) = \frac{P(\theta)P(D|\theta)}{P(D)}$$

$\swarrow$ Posterior distribution of θ

$\searrow$ Normalizing constant (makes it a probability distribution)

“our current knowledge is a mixture of what we previously knew updated with new data”

$$P(D) = \int P(\theta)P(D|\theta)d\theta$$

$$P(\theta|D) = \frac{P(\theta)P(D|\theta)}{\int P(\theta)P(D|\theta)d\theta} \longrightarrow \text{“integrating over the marginals”}$$

A BAYESIAN VIEW

Some extra definitions

	τ_A	τ_B	τ_C	$P(V^A; T^c)$
v^A	0.10	0.07	0.12	0.29
v^B	0.05	0.22	0.06	0.33
v^C	0.05	0.19	0.14	0.38
	0.20	0.48	0.32	

Marginal probabilities

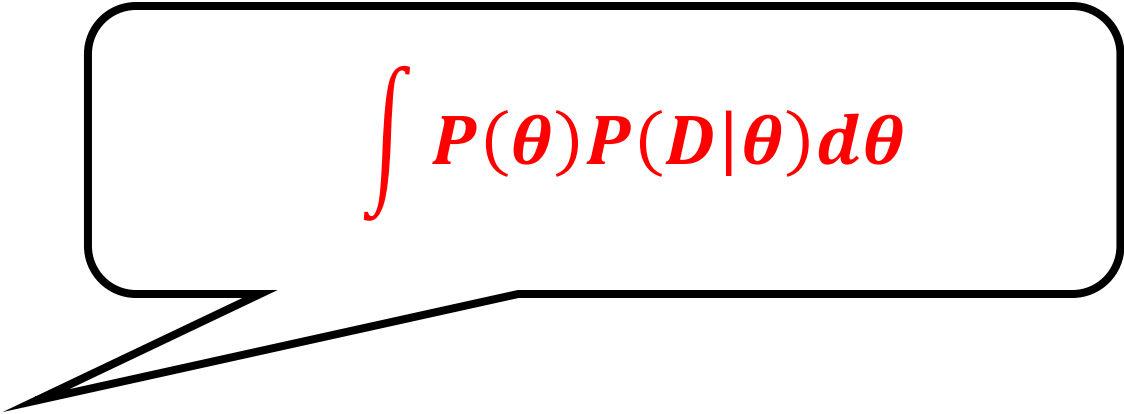
Implementing Bayesian

$$P(\theta|D) = \frac{P(\theta)P(D|\theta)}{P(D)} = \frac{P(\theta)P(D|\theta)}{\int P(\theta)P(D|\theta)d\theta}$$

$$\int P(\theta)P(D|\theta)d\theta$$

Implementing Bayesian

$$P(\theta|D) = \frac{P(\theta)P(D|\theta)}{P(D)} = \frac{P(\theta)P(D|\theta)}{\int P(\theta)P(D|\theta)d\theta}$$


$$\int P(\theta)P(D|\theta)d\theta$$

What if it does not exist?

Implementing Bayesian

SOLUTION MCMC

$$P(\theta|D) = \frac{P(\theta)P(D|\theta)}{P(D)} = \frac{P(\theta)P(D|\theta)}{\int P(\theta)P(D|\theta)d\theta}$$

$$\int P(\theta)P(D|\theta)d\theta$$

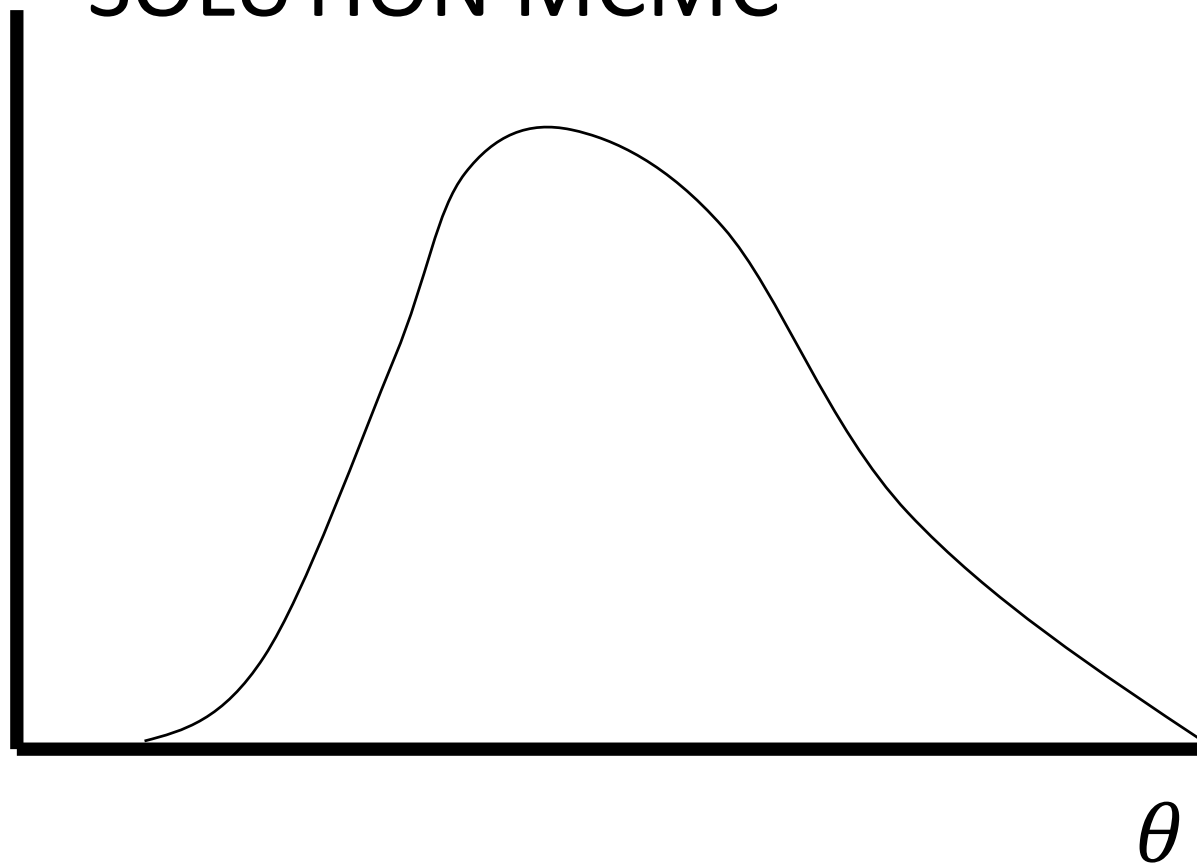
What if it does not exist?

What if there is no conjugate?

Implementing Bayesian

SOLUTION MCMC

$$\frac{P(\theta|D)P(\theta)}{P(D)}$$



Implementing Bayesian

SOLUTION MCMC

$$\frac{P(\theta|D)P(\theta)}{P(D)}$$

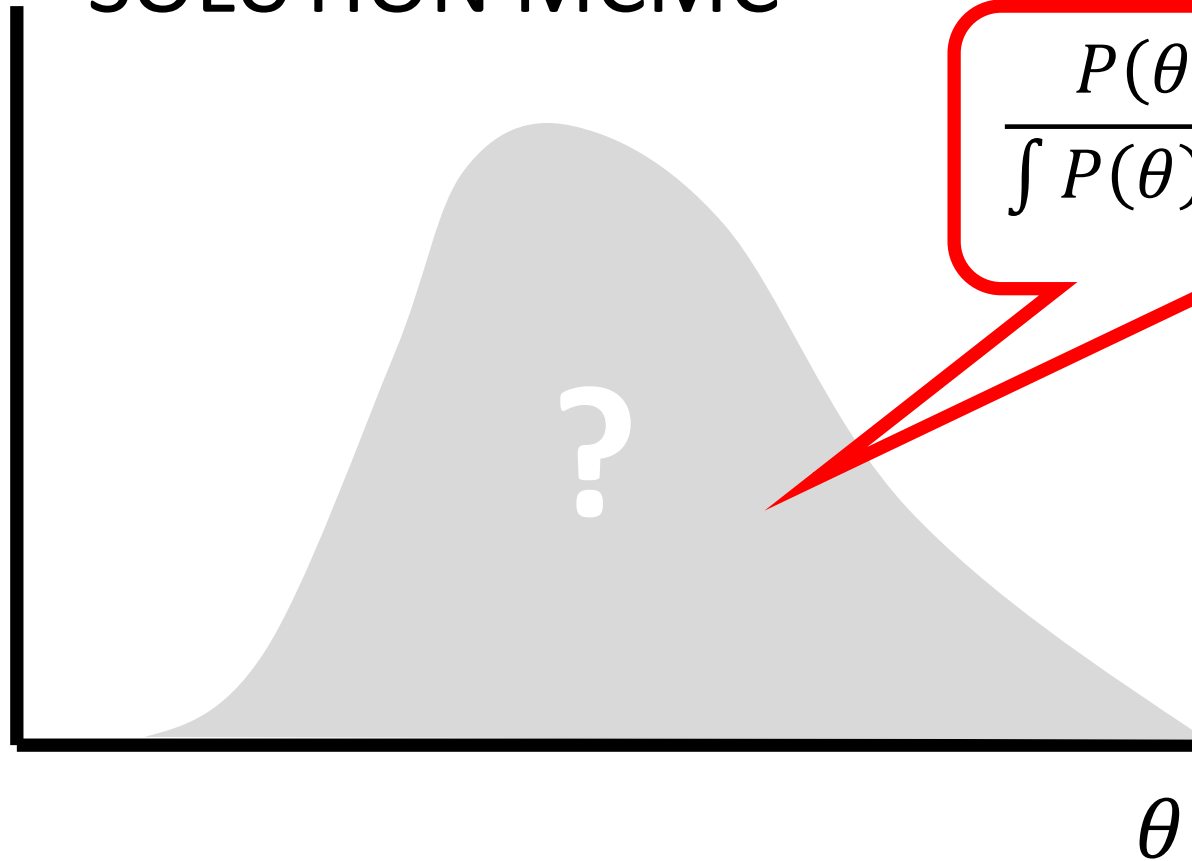
At least, is there any way we can generate samples from this (unknown) posterior distribution so we can broadly regenerate the (unknown) posterior distribution?

θ

Implementing Bayesian

SOLUTION MCMC

$$\frac{P(\theta|D)P(\theta)}{P(D)}$$



Implementing Bayesian

SOLUTION MCMC

$$\frac{P(\theta|D)P(\theta)}{P(D)}$$



$$\frac{P(\theta)P(D|\theta)}{K}$$

Implementing Bayesian

SOLUTION MCMC

How many times is more likely j than i in this (unknown) distribution?

$$r(i, j) = \frac{\frac{P(\theta = i)P(D|\theta = i)}{K}}{\frac{P(\theta = j)P(D|\theta = j)}{K}} = \frac{P(\theta = i)P(D|\theta = i)}{P(\theta = j)P(D|\theta = j)} = \frac{\pi(j)}{\pi(i)}$$

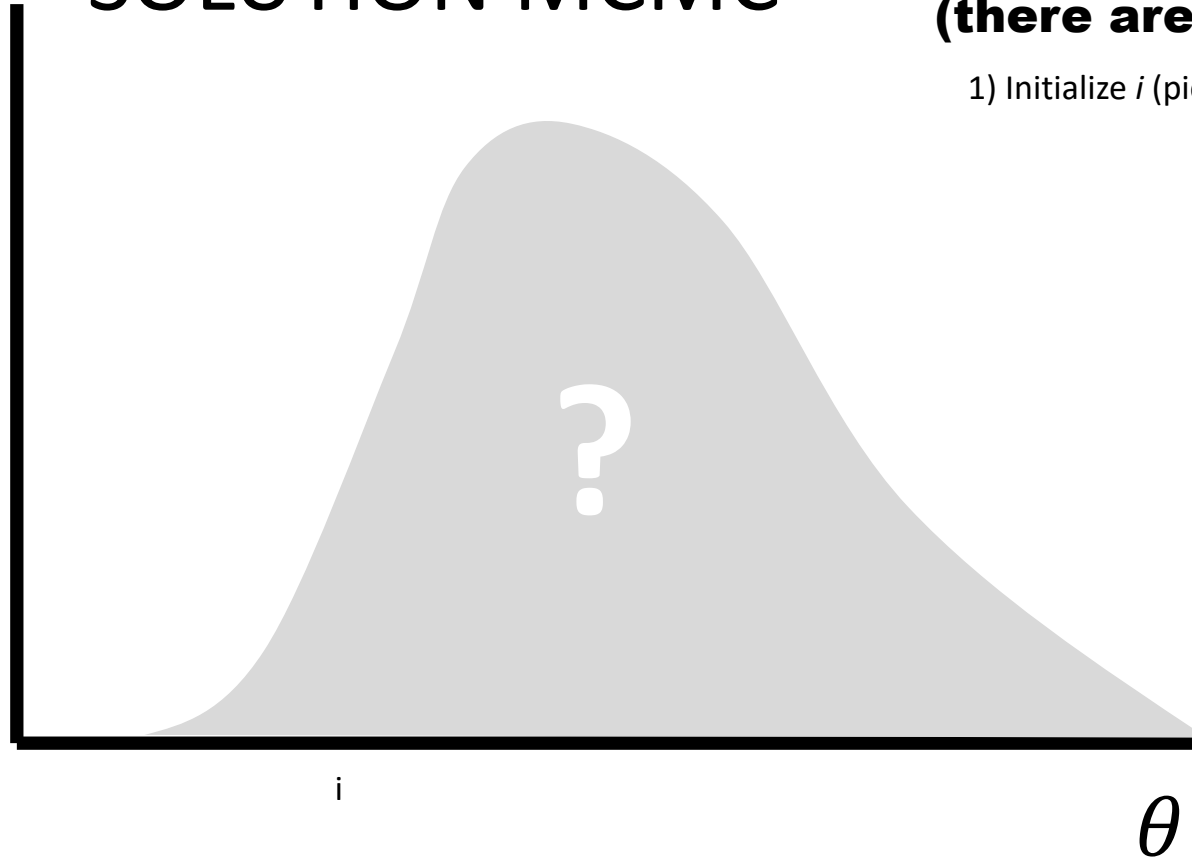
Implementing Bayesian

SOLUTION MCMC

**Metropolis algorithm
(there are many others!)**

1) Initialize i (pick a value of θ at random)

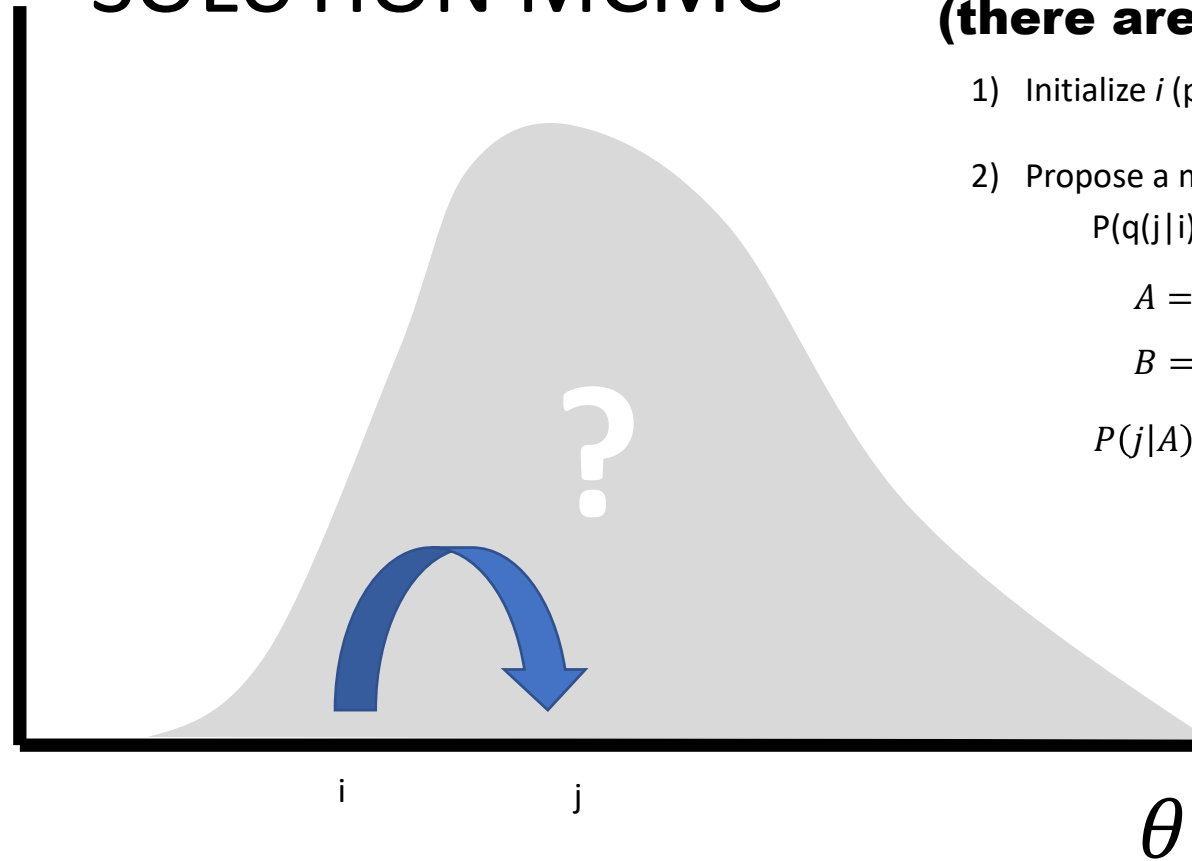
$$\frac{P(\theta|D)P(\theta)}{P(D)}$$



Implementing Bayesian

SOLUTION MCMC

$$\frac{P(\theta|D)P(\theta)}{P(D)}$$



Metropolis algorithm (there are many others!)

1) Initialize i (pick a value of θ at random)

2) Propose a movement $q(\cdot|i)$

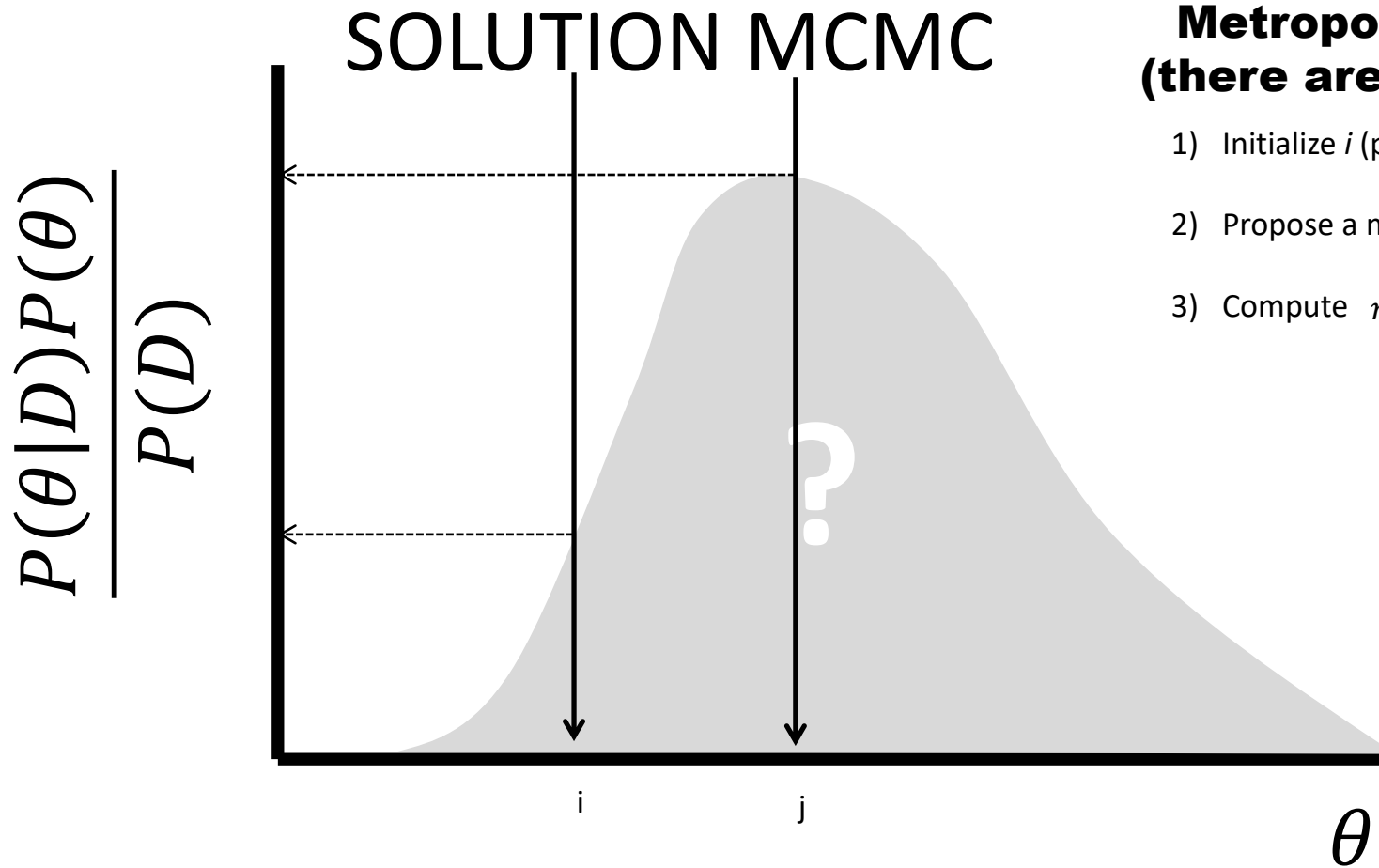
$$P(q(j|i))=P(q(i|j))$$

$$A = N(i, \sigma)$$

$$B = N(j, \sigma)$$

$$P(j|A) = p(i|B)$$

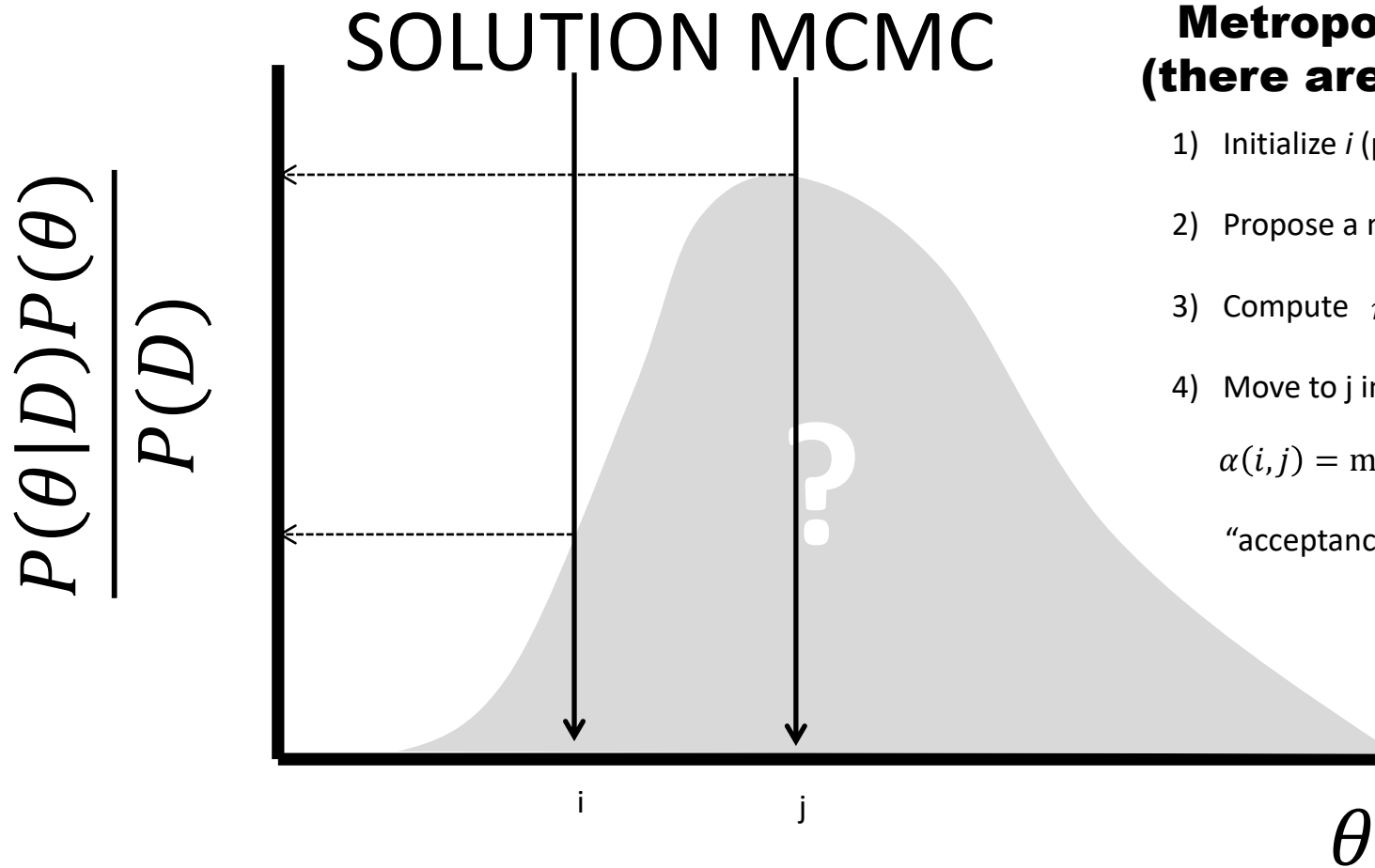
Implementing Bayesian



Metropolis algorithm (there are many others!)

- 1) Initialize i (pick a value of θ at random)
- 2) Propose a movement $q(\cdot | i)$
- 3) Compute $r(i, j) = \frac{\pi(j)}{\pi(i)}$

Implementing Bayesian



Metropolis algorithm (there are many others!)

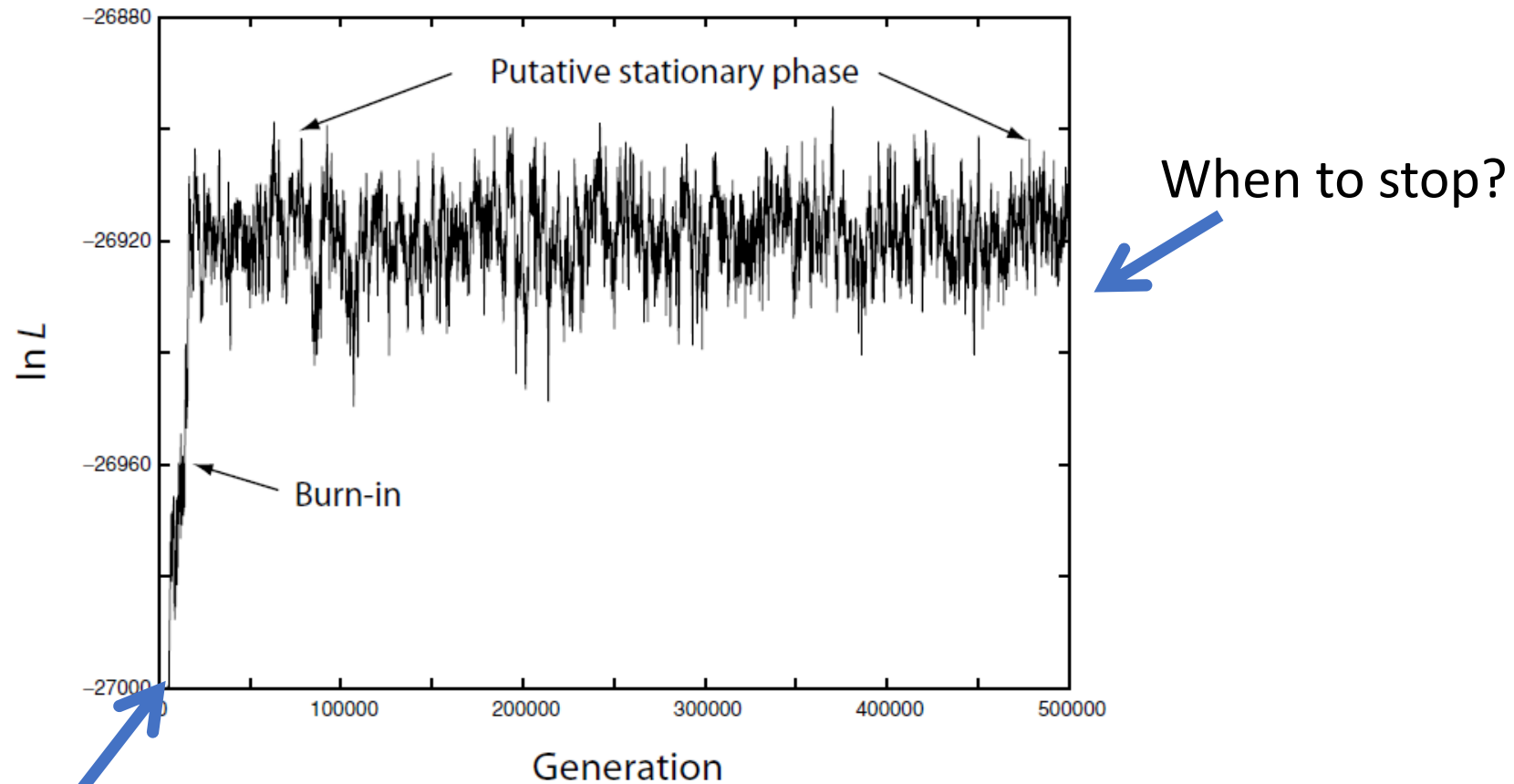
- 1) Initialize i (pick a value of θ at random)
- 2) Propose a movement $q(\cdot | i)$
- 3) Compute $r(i, j) = \frac{\pi(j)}{\pi(i)}$
- 4) Move to j in probability

$$\alpha(i, j) = \min(1, r(i, j))$$

“acceptance ratio”

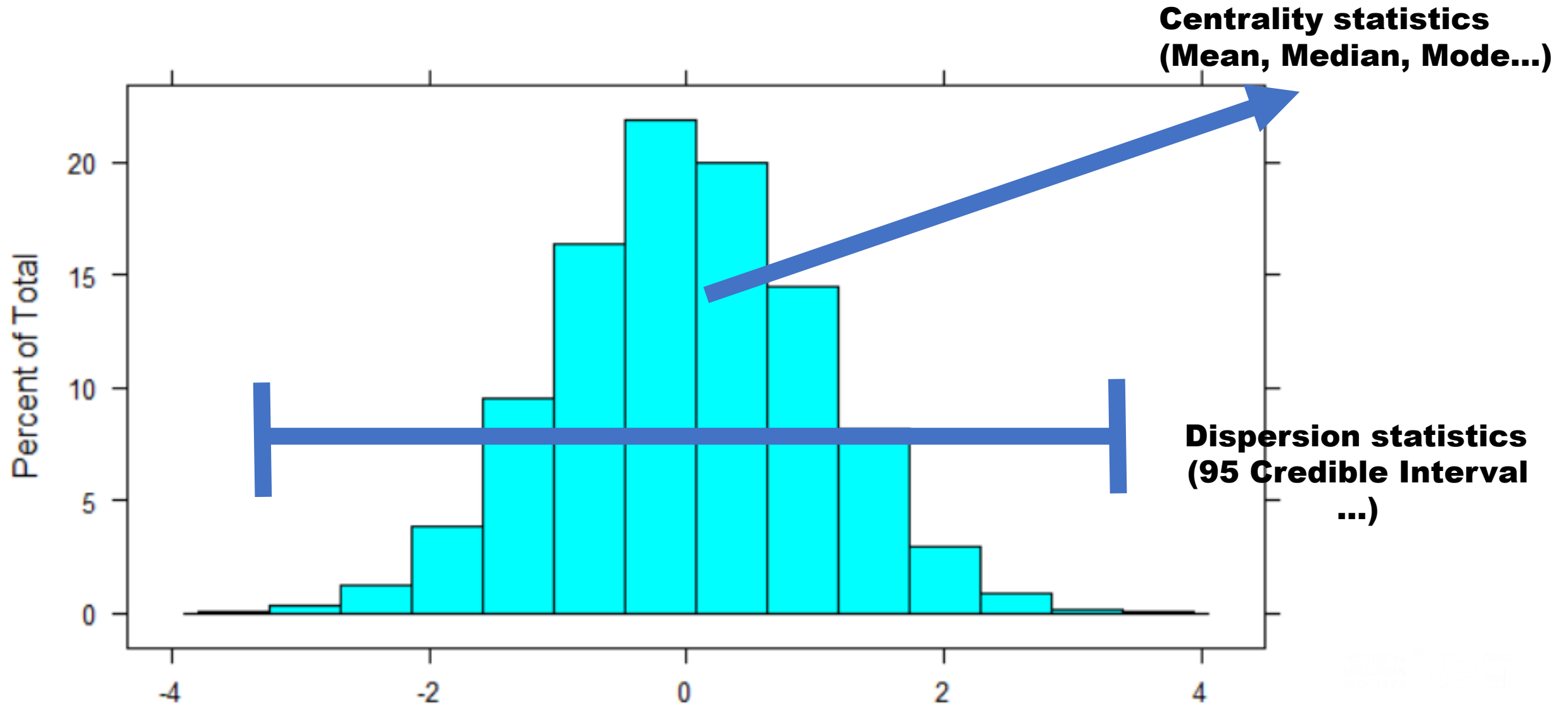
Implementing Bayesian

SOLUTION MCMC



Usually we start out of the most likely values of the posterior

Implementing Bayesian



Implementing Bayesian

*“**Model adequacy** is a different concern than model fit: the latter determines which of a set of models is least bad (or “best”) for the data, whereas the former determines whether a single model adequately describes the data. For example, a model that says humans, chimps, and mice all diverged simultaneously 70 million years ago is a better fit to the data than one that puts that divergence 700 million years ago, but neither model adequately describes the data. Adequacy is often evaluated by simulating under the focal model to see if it generates data indistinguishable from the empirical data using one or more measures.”*

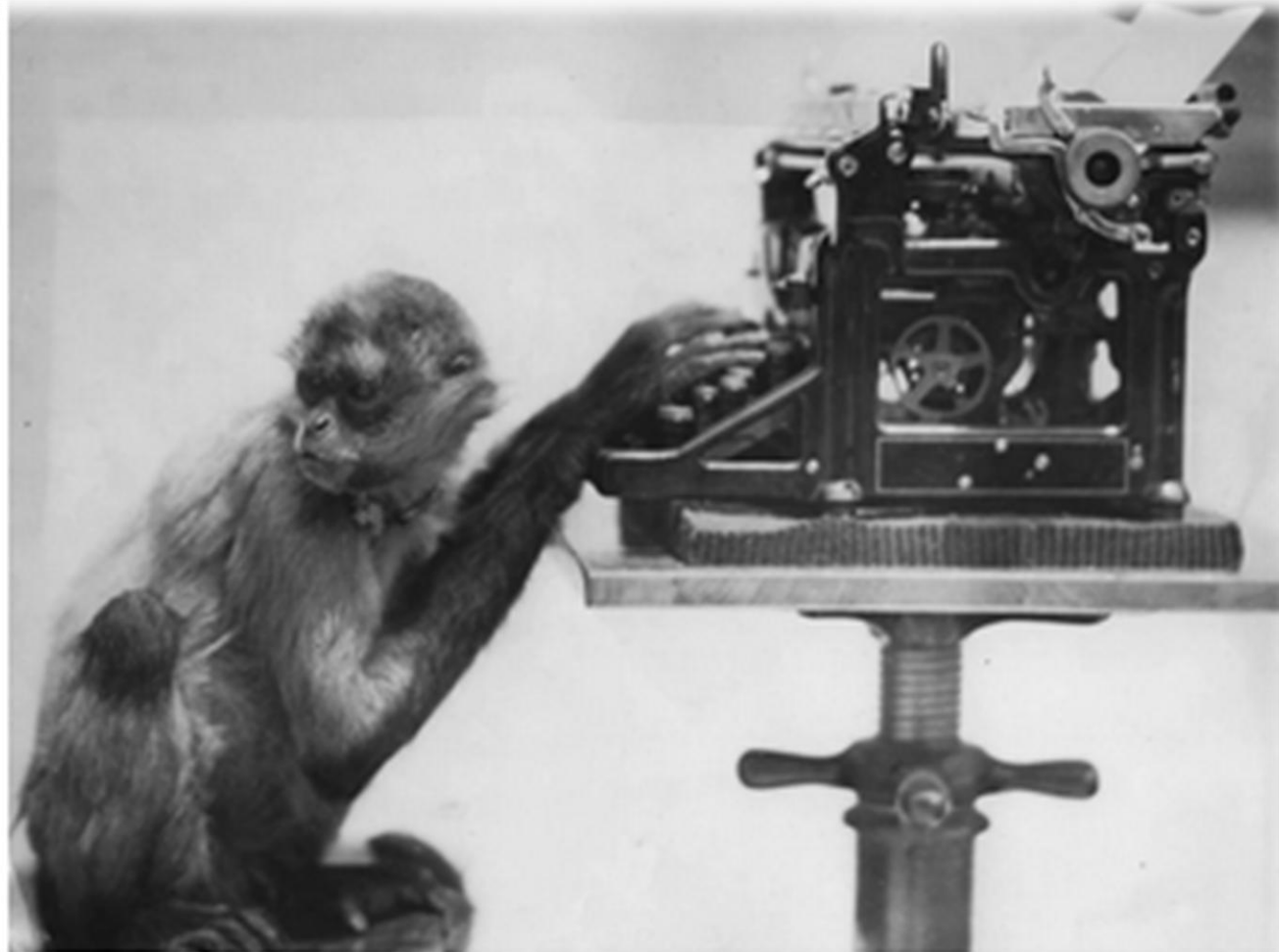
Implementing Bayesian

In practice: sample from the posterior distribution, generate datasets, and check out how similar are these simulated datasets to the observed data

Exercise 8

- Do Metropolis MCMC algorithm

IT IS YOUR RESPONSIBILITY!



Thank you very much!

